

Infectious Diseases

November 30, 2025

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Overview



1. Microbiology 101
2. CNS Infections
3. CV Infections (IE)
4. Pulmonary Infections
5. Infectious Diarrhea
6. Intraabdominal Infections
7. GU Infections (UTIs) and 8. STIs
9. Skin and Soft Tissue (SSTIs)
10. Bone & Joint Infections (OM)
11. HIV
12. TB and NTM
13. Travel & Tropical & Parasitic Infections
14. Fungal Infection

BONUS SLIDES IN YOUR COURSEPACK **("easy reading!")**

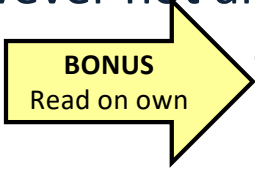
- A) Emerging Infections
- B) Tick-Borne Infection
- C) Fever of Unknown Origin (FUO)
- D) Immunocompromised Host
- E) Nosocomial infections
- F) Infection Control
- G) Exposure Management (Bites, needlestick)
- H) Adult Immunizations
- I) Choosing Wisely Canada ID recommendations
- J) Bonus Questions



= recent guideline or landmark article or high yield!

ID Lecture Notes

- Many potential guidelines
 - Be aware of Canadian vs. American guidelines. If there are Canadian ones generally these should be followed.
 - When using American (IDSA) guidelines be aware that our resistance profile is very different, e.g. much lower rates of MRSA generally
- Lecture Organization regarding BONUS slides
 - Slides are grouped by topic, however not all slides will be covered by speaker. If you see a slide with skip over it to stay on time but slide prior to exam!



BONUS
Read on own

on it, Dr Verway is going to be sure to go back and read



Acronyms Used in this Talk

- AG = Aminoglycoside
- ARO = Antimicrobial resistant organism
- ART/ARV = antiretroviral therapy/antiretrovirals
- C+G = Chlamydia and Gonorrhea
- CAUTI = Catheter-associated urinary tract infection
- CLABSI = Central line associated bloodstream infection
- CNST/CONS = Coagulase negative Staphylococcus
- CMIA = Chemiluminescent microparticle immunoassay
- CPIS = Clinical Pulmonary Infection Score
- CPE = Carbapenemase producing Enterobacterales
- CRE = Carbapenem resistant Enterobacterales
- ESBL = Extended Spectrum Beta Lactamase
- EIA = Enzyme immunoassay
- FN = Febrile Neutropenia
- FQ = Fluoroquinolone
- HAP/VAP = Hospital Associated Pneumonia/Ventilator Associated Pneumonia
- HBO = Hyperbaric oxygen
- HCW = Health Care Worker
- HSCT: Hematopoietic stem cell transplant
- IC = Immunocompromised
- IMD = Invasive Meningococcal Disease
- INSTI = Integrase inhibitor
- LN = Lymphadenopathy/lymph nodes
- MS = Mental Status
- NNRTI/NRTI = (Non) Nucleoside reverse transcriptase inhibitor
- OI = Opportunistic infection
- OM = Osteomyelitis
- OMM = Optimal medical management
- PI = Protease inhibitor
- PID = Primary immunodeficiency
- PJI = Prosthetic joint infection
- PsA = Pseudomonas aeruginosa
- PWID = People who inject drugs
- RIPE = Rifampin, Isoniazid, Pyrazinamide, Ethambutol
- RPR: Rapid plasma reagin
- SOT = Solid organ transplant
- STEC = Shiga-toxin producing *Escherichia coli*
- TMP-SMX = Trimethoprim-sulfamethoxazole
- TOC = Test of cure
- TPPA = T. pallidum particle agglutination
- ULN = Upper limit of normal

1. Microbiology 101

References:

- Bennett, John E. (John Eugene), Raphael Dolin, and Martin J. Blaser, eds. Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases. Ninth edition. Philadelphia, PA: Elsevier, 2020.
- Pranita D Tamma, Emily L. Heil, Julie Ann Justo, Amy J. Mathers, Michael J. Satlin, & Robert A. Bonomo. IDSA 2024 Guidance on the Treatment of Antimicrobial Resistant Gram-Negative Infections. Infectious Diseases Society of America July 2024. Available at <https://www.idsociety.org/practice-guideline/amr-guidance/>.

Bacteriology 101

	GRAM POSITIVE ("purple")	GRAM NEGATIVE ("pink")
Cocci	• Clusters – <i>S. aureus</i> or CNST (Coagulase Negative Staph) • Pairs/chains – <i>Streptococcus spp.</i> or <i>Enterococcus</i>	• <i>Neisseria meningitidis/gonorrhea</i> (diplococci) • <i>Moraxella spp.</i>
Bacilli (rods)	• Spore forming – <i>Bacillus</i> or <i>Clostridium</i> • Non-spore forming – <i>Corynebacterium</i>, <i>Listeria</i> • Anaerobes: <i>Cutibacterium</i>, <i>Actinomyces</i> (branching)	• Lactose fermenters – <i>Klebsiella</i>, <i>E. coli</i>, <i>Enterobacter</i>, <i>Proteus</i> etc (gut bugs) • Non-lactose fermenters – <i>Pseudomonas</i>, <i>Stenotrophomonas</i> • HACEK organisms • Anaerobes: <i>Bacteroides</i>, <i>Fusobacterium</i>

Spirochetes (spirals)

Treponema spp, *Leptospirosis*, *Borrelia (Lyme)*

Mycobacteria

M. tuberculosis, *MAC*, *M. leprae*

Fungi

- **Yeast** (ie. *Candida* spp., *Cryptococcus* spp.)
 - Asexual, unicellular organisms
 - Typically superficial infection but invasive disease in immunocompromised hosts
- **Mold** (ie. *Aspergillus* spp., *Mucor* spp.)
 - Multicellular organisms that have hyphae
- **Dimorphic fungi** (ie. *Blastomycosis* spp. *Histoplasmosis* spp., *Coccidiomycosis* spp.)
 - Exist as yeast in body temperature but becomes mold in room temperature, “mold in the cold, yeast in the beast”

Parasites (non-exhaustive)

Parasitology is historically low-yield on RC Exam!

Protozoa

Amoebae

Acanthamoeba
Naegleria fowleri
Entamoeba histolytica

Apicomplexans

Plasmodium
Babesia
Toxoplasma
Cryptosporidium

Flagellates

Trypanosoma
Leishmania
Giardia
Dientamoeba

Helminths

Flat Worms (Platyhelminths)

Flukes (Trematodes)

Schistosoma
Fasciola
Paragonimus
Opisthorchis
Clonorchis

Tapeworm (Cestodes)

Echinococcus
Taenia solium
Taenia saginata
Diphyllobothrium

Roundworm (Nematodes)

Ascaris
Strongyloides
Trichinella
Trichuria
Enterobius

Microfilaria
W. bancrofti
Onchocerca



High Yield Acronyms/Antibiotics

ESBL – eg *E. coli*, *Klebsiella* **Rx:** Carbapenem, TMP-SMX, FQ, AG (if sensitive)

CRE Rx: Depends on CRE mechanism and sensitivities! Call ID for new generation antibiotics and guidance

MRSA Rx: Vancomycin, Doxycycline, TMP-SMX, Clindamycin, Linezolid, Daptomycin, Ceftobiprole

Pseudomonas Rx: Pip/Tazo, Ceftazidime, Cefepime, Imipenem, Meropenem, Ciprofloxacin, AG, Aztreonam, Colistin, Tigecycline, Ceftazidime-Avibactam, Ceftolozane-tazobactam

Enterococcus Rx: Ampicillin (if S) or Vanco (not VRE), Linezolid, Daptomycin

Broad Spectrum B-Lactamases

ESBL	Amp C
Class A	Class C
Plasmid Mediated	Chromosome Mediated
Constitutive Resistance	Inducible Resistance**

**variable risk of developing resistance in presence of beta-lactam, they may be “sensitive” at time of culture, but resistance may develop after a few days on treatment

AmpC Organisms:

- **OLD acronym - SPICE HAM** – *Serratia*, *Providencia*, *Indole-positive Proteus*, *Citrobacter*, *Enterobacter*, *Hafnia*, *Acinetobacter*, *Morganella*
- **NEW acronym - ECK!** - *Enterobacter cloacae*, *Citrobacter freundii* (but not koseri), *Klebsiella aerogenes*



Rx: Carbapenem, TMP-SMX, Fluoroquinolone (FQ), Aminoglycoside (AG)

2. CNS Infections

References:



- Tunkel AR, Hartman BJ, Kaplan SL, *et al.* Practice guidelines for the management of bacterial meningitis. *Clin Infect Dis* 2004;39:1267-84.
- Attia J, Hatala R, Cook DJ, and Wong JG. Does this adult patient have acute meningitis? *JAMA* 1999;282:175-81.
- Brouwer MC, McIntyre P, Prasad K, and van de Beek D. Corticosteroids for acute bacterial meningitis. *Cochrane Database of Systematic Reviews* 2015, Issue 9. Art No: CD004405.
- Public Health Agency of Canada. Canadian Immunization Guide: Part 4 – Active Vaccines; Meningococcal Vaccine. PHAC; Ottawa (ON): PHAC; [updated 2020].
- Tunkel AR *et al.* 2017 IDSA Guidelines for Healthcare associated ventriculitis/meningitis. *Clin Infect Dis*. 2017; 64:e34-65.

CNS Infections

Meningitis

Predominantly starts with headache, neck stiffness and fever and can get altered LOC later into course.

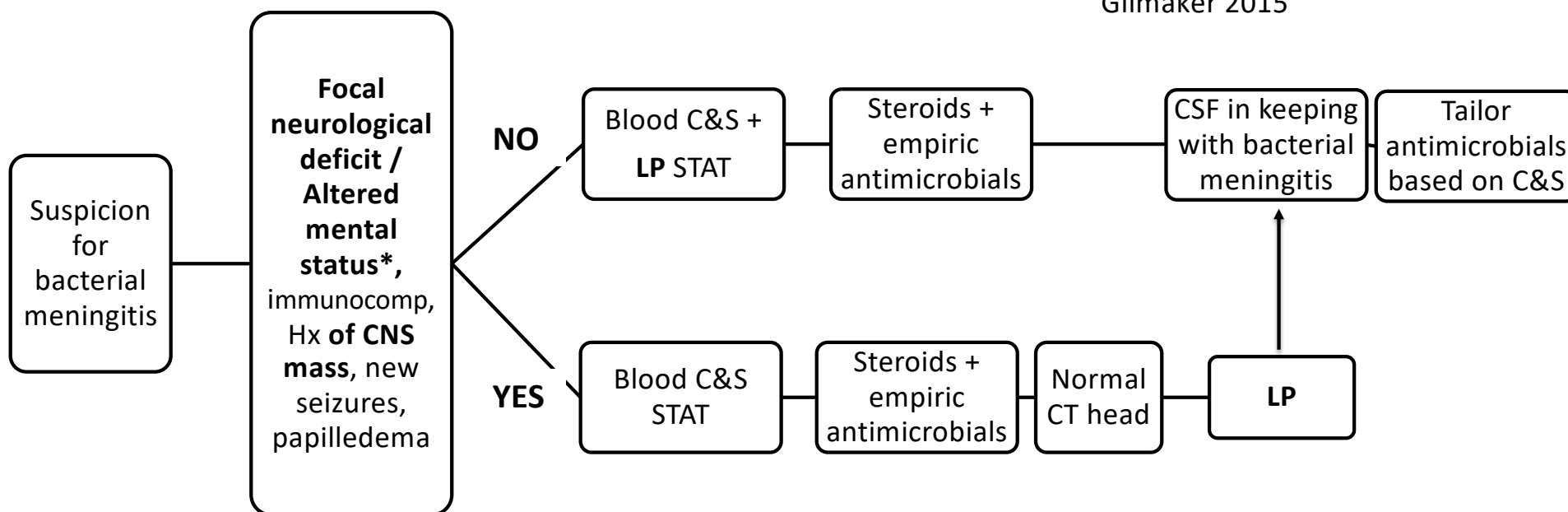
Encephalitis

Predominantly starts with altered LOC/mental status and fever and can get seizures, focal neurological changes associated.

- **JAMA: Does this adult patient have acute meningitis?**
 - Meningitis ruled out (99%) if fever, neck stiffness and altered MS all absent
 - Jolt accentuation – high sensitivity (97% in one study)
 - Kernig's and Brudzinski's signs – high specificity, poor sensitivity
- **Basal skull meningitis:**
 - + CN palsies, long-tract signs
 - Think TB, *Listeria*, *Cryptococcus*, Syphilis, Lyme in correct host

Bacterial Meningitis – CT or no CT?

Tunkel 2004
Glimåker 2015



Meningitis – CSF Parameters

	NORMAL	BACTERIAL	VIRAL	TB	FUNGAL
Opening Pressure (cmH2O)	5-20	>30	Normal	May be Increased	May be increased
WBC (cells/mL)	<5	> 1000	< 1000	Variable	Variable
Differential		Neutrophils	Lymphocytes	Lymphocytes	Lymphocytes
Glucose (mmol/L)	2.5-3.5	↓ (<2.2)	N	↓	↓
Protein(g/L)	0.18-0.45	↑ (>1)	N to ↑	↑	↑

- Differential may be unreliable in recent symptom onset
- CSF biochemistry & cell count minimally affected at 2 days by antibiotics
- Individual CSF predictors for bacterial meningitis each with > 99% certainty
 - Glucose < 1.9 mmol/L, CSF:blood glucose < 0.23, protein > 2.2 g/L, WBC > 2000 cells/mL, PMNs > 1180 cells/mL

Meningitis – Pathogens + Rx

Predisposing factor	Common bacterial pathogens	Antimicrobial Rx - Empiric
18-50 years	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>H. influenzae</i>	Vancomycin + Ceftriaxone
> 50 years or immunocompromised	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>H. influenzae</i> , <i>L. monocytogenes</i>	Vancomycin + Ceftriaxone + Ampicillin

***In case of PCN allergy, use Vancomycin + Moxifloxacin +/- TMP-SMX (if *Listeria* coverage needed)
Empiric VANCO = 15-20mg/kg IV q12h, consult pharmacy for dose adjustments

CSF Gram stains:

Gram-positive diplococci → *S. pneumoniae*

Gram-negative diplococci → *N. meningitidis*

Gram-negative bacilli or coccobacilli → *H. flu*

Gram-positive bacilli → *L. monocytogenes*

Bug	Drug	Duration
<i>S. pneumoniae</i>	Pen G 4MU IV q4h or CTX 2g IV q12h	10-14 days Abx DEX x 4 days
<i>N. meningitidis</i> <i>H. flu</i>	CTX 2g IV q12h	4-7 days
<i>L. monocytogenes</i>	Ampicillin 2g IV q4h	21 days

Meningitis - Steroids

- Dexamethasone 0.15 mg/kg IV q6h for 4 days PRIOR TO or WITH first dose of antibiotics
 - 50% ↓ in mortality/morbidity for pneumococcal meningitis in one large RCT
 - Cochrane Review (2015): Steroids associated with
 - ↓ in mortality for meningitis secondary to *Streptococcus pneumoniae* (29.9 vs 36%)
 - ↓ in hearing loss and short-term neurological sequelae in high income countries only for all pathogens
 - May have increased mortality in neurolisteriosis
- **General approach:**
 - Administer before or with first dose of empiric antibiotics for suspected bacterial meningitis
 - Stop if CSF is non-turbid OR low cell count OR non-pneumococcal by culture
 - Do not start if antibiotics have already been given to patient

Brower et al, 2015

Gans JD et al, 2002

Charlier et al, 2017



Neisseria meningitis – **CHEMO**prophylaxis

- Who?
 - **Household contacts**
 - Persons **sharing sleeping arrangements**
 - Persons who have direct nose/mouth contamination w oral/nasal secretions
 - Children and staff in childcare or nursery
 - **HCWs who have had intensive unprotected contact** (without wearing a mask) (eg. **intubating, resuscitating, closely examining oropharynx**)
 - Airline passengers sitting immediately on either side of the case (but not across the aisle) when total time on aircraft ≥ 8 hours
- When?
 - Consider risk of exposure if they had contact 7 days before case showed symptoms until 24h post initiation of treatment for case
 - Initiate treatment for contacts within 10 days, ideally within first 24h of identification of the case
- What?
 - Ciprofloxacin 500 mg PO X 1 dose (increasing resistance concern)
 - OR ceftriaxone 250 mg IM X 1 dose
 - OR rifampin 600 mg PO BID X 2d



N. meningitis – IMMUNOprophylaxis

- **Who?**
 - **Household contacts** of a case of invasive meningococcal disease (IMD)
 - Persons **sharing sleeping arrangements** with a case of IMD
 - Persons who have direct nose/mouth contamination with oral/nasal secretions of a case with IMD
 - Children and staff in contact with a case of IMD in childcare or nursery school facilities
- **What?**
 - Depends on serotype of index case and age / underlying conditions of contact
 - Men-C-ACYW or 4CMenB can be considered



Oral Exam?

“I will notify public health of case of meningococcal meningitis ...”

(In truth **all lab confirmed cases of bacterial and viral meningitis are reported by lab to Public Health Ontario automatically**, but here you want PH to do contact tracing)

Healthcare–Associated Ventriculitis & Meningitis

BONUS

Read on own

Signs and Symptoms

- New headache, nausea, lethargy, altered LOC
- Erythema and pain over SC shunt tubing

Investigations

- Blood cultures, CSF culture including fungal culture (hold for 10 days)

Management

1. Complete removal of infected CSF shunt / hardware, replaced with external drain
2. Empiric Treatment: Vancomycin PLUS Ceftazidime OR Meropenem
 - +/- Rifampin (if staphylococcal isolate)
 - +/- intraventricular Abx if no response to systemic Abx
3. Repeat CSF cultures to confirm negative growth
 - Duration of therapy 10-14 (gram positives) up to 21 days (gram negatives) from last positive culture
4. Reimplantation of new shunt once repeat CSF culture have been negative for 7-10 days

3. CV Infections: Infective Endocarditis

References:



- Baddour LM, Wilson WR, Bayer AS, *et al.* Infective endocarditis in adults: diagnosis, antimicrobial therapy and management of complications: a scientific statement for healthcare professionals from the American Heart Association. *Circulation*. 2015;132:1-52.
- Otto CM, Nishimura RA, Bonow RO, *et al.* 2020 ACC/AHA Guideline for the management of patients with Valvular Heart Disease. *Circulation*. 2021; 143.
- Wilson WR, Gewitz M, Lockhart PB *et al.* Prevention of viridans group streptococcal infective endocarditis: A scientific statement from the American Heart Association. *Circulation*. 2021;143:e963-e978.
- Wilson W, Taubert KA, Gewitz M, *et al.* Prevention of infective endocarditis: guidelines from the American Heart Association. *Circulation*. 2007;116(15):1736-1754.
- Fowler VG, Durack DT, Selton-Suty C, *et al.* The 2023 Duke-International Society for Cardiovascular Infectious Diseases Criteria for Infective Endocarditis: Updating the Modified Duke Criteria [published correction appears in Clin Infect Dis. 2023 Oct 13;77(8):1222]. *Clin Infect Dis*. 2023;77(4):518-526. doi:10.1093/cid/ciad271



IE Workup –Class I Recommendations (AHA 2021)

- Diagnostic workup
 - at least 2 sets of blood cultures prior to antibiotics
 - Initial TTE for everyone.
 - TEE Class I indications: TTE nondiagnostic or IE complications suspected or intracardiac leads (TEE widely used these are just the Class I – eg consider if staphylococcal, enterococcal, fungal infections)
- In patients being considered for an early change to oral antibiotic therapy for treatment of stable IE, baseline TEE before switching to Oral and repeat TEE 1-3 days before completing antibiotic regimen should be performed
- Patients with suspected or confirmed IE related to IV drug use should be referred to addictions medicine specialist for opioid substitution therapy
 - See Critical Care/Toxicology lecture for overview of Opioid Agonist Therapy

IE – Diagnosis (New Duke Criteria in 2023! changes italicized) *Fowler VG et al, 2023*

Major Criteria (M)

A) Microbiologic Criteria (≥1 of):

- ≥2 BCx with typical organisms:
 - *S. aureus/lugdunensis, Streptococci (except GAS or pneumo), E. faecalis, HACEK, Granulicatella, Abiotrophia, Gemella in Native Valve*
- ≥3 BCx with occasional/rare organisms
- *Blood PCR for Coxiella, Bartonella, T. whipplei*
- *Coxiella burnetii* in 1 BCx or IgG ≥ 1:800
- *Bartonella henselae or quintana* IgG ≥ 1:800

B) Imaging Criteria

- Echo or Cardiac CT: vegetation, valve/leaflet perforation or aneurysm, abscess, pseudoaneurysm, fistula, prosthetic valve dehiscence or New significant valvular regurgitation
- PET: Abnormal activity of valve, aortic graft, or intracardiac leads

C) Surgical Criteria

- Evidence of IE by direct inspection during surgery

Minor Criteria (m)

Predisposition: Prior IE, Prosthetic Valve, Prior Valve Repair, Congenital heart disease, Regurgitation/Stenosis, Pacemaker/ICD, HOCM, PWID

Fever: Documented temp >38°C

Vascular Phenomena: Arterial emboli, septic pulmonary infarcts, cerebral or splenic abscess, mycotic aneurysm, ICH, conjunctival hemorrhage, Janeway lesion, purulent purpura

Immunologic Phenomena: Rheumatoid Factor, Osler Nodes, Roth Spots, immune-complex GN

Microbiologic Evidence: BCx for organism consistent with IE not meeting major criteria, *BCx/PCR with organism consistent with IE from sterile body site, PCR for skin bacteria on valve/wire*

Imaging Criteria: Abnormal activity on PET within 3m of implant of prosthetic valve, aortic graft, intracardiac leads, prosthetic material

Physical Exam Criteria: *New* Valve regurgitation on auscultation

DEFINITE IE

- Microorganism identified on vegetation/valve
- 2M OR 1M+3m OR 5m criteria

POSSIBLE IE

- 1M+1m OR 3m criteria

REJECTED IE

- Alternative Diagnosis
- Lack of recurrence with <4d Abx
- Negative pathology with <4d Abx
- Does not meet Possible IE criteria

IE – Antibiotic Treatment

	Native Valve	Prosthetic Valve
MSSA	Cloxacillin OR cefazolin	Cloxacillin OR cefazolin <u>PLUS</u> rifampin <u>PLUS</u> gentamicin
MRSA or CNST [^]	Vancomycin	Vanco <u>PLUS</u> rifampin <u>PLUS</u> gentamicin
Viridans group strep, <i>S. gallolyticus/bovis</i>	PenG* OR ceftriaxone*	
<i>Enterococcus faecalis</i>	Ampicillin PLUS gentamicin OR Ampicillin PLUS ceftriaxone	
<i>Enterococcus faecium</i>	Vancomycin PLUS gentamicin	
HACEK group	Ceftriaxone	

Duration:

- 4-6 weeks**

[^]Most (~85%) CNST are considered beta-lactam resistant, but if susceptible can treat like MSSA

* Consider adding aminoglycoside with higher MIC to PCN levels.

** VGS duration may range from 2-6 weeks depending on MIC



IE – CLASS I Surgical Indications

****2021 AHA/ACC Guideline****

The most important indication

Decisions about surgery should be made by a (Class I Level B)
“multispecialty heart valve team” of cardiologists,
cardiovascular surgeons and Infectious Disease
Specialists



IE – CLASS I Surgical Indications

** 2021 AHA/ACC Guideline**

Early Surgical Indications - Class I indications

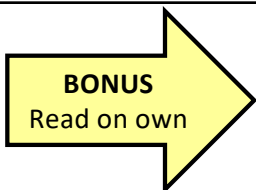
“Early Surgery = during initial hospitalization before full treatment course of antibiotics is completed”

STRENGTH OF EVIDENCE

Valve dysfunction with signs or symptoms of heart failure (persistent despite OMM)	Class I, Level B
Left-sided IE caused by <i>S. aureus</i> , fungi or highly resistant organisms	Class I, Level B
Heart block, annular/aortic root abscess, destructive penetrating lesions	Class I, Level B
Persistent bacteremia or fever > 5d after starting appropriate antibiotics	Class I, Level B
Complete removal of implantable electronic cardiac device (PPM/CRT/ICD) systems in patients with definite endocarditis	Class I, Level B

Other Surgery Indication:

PROSTHETIC Valves: Relapsing infection (defined as new fevers/bacteremia after a complete course of appropriate antibiotics and interval sterile blood cultures, no other source/portal for infection)	Class I, Level C
In patients with recurrent endocarditis in the setting of addictions and injection drug use, consultation with addiction medicine is recommended ... before repeat surgical intervention is considered	Class I, Level C



IE – Class II Surgical Indications

**** 2021 AHA/ACC Guideline****

Class II Other Surgical Indications Endocarditis	STRENGTH
Early surgery (during initial hospitalization and before completion of course of antibiotics) is reasonable in those with recurrent emboli and persistent vegetations despite appropriate antimicrobial therapy	Class IIa, Level B
Early surgery (during initial hospitalization and before completion of course of antibiotics) may be considered in patients with Native Left sided valvular endocarditis with mobile vegetation > 10mm With or WITHOUT embolic phenomenon , esp when anterior MV leaflet	Class IIb, Level B
Early (as opposed to delayed) surgery for patients who have had a minor (no extensive neuro deficits) embolic stroke without ICH, in patients with an indication for IE surgery	Class IIb, level B
Consider delay IE surgery >4 weeks in hemodynamically stable patients after a major ischemic or hemorrhagic stroke.	Class IIb, Level B

IE - Prophylaxis

Wilson et al, 2021
AHA/ACC Valve 2020 Guideline
Wilson et al, 2007

Patient Population	Procedure
Prosthetic cardiac valve (including TAVI, devices e.g. annuloplasty ring, clips, LVAD) <i>*Not indicated for PPM/ICD, coronary stent</i>	Dental procedures involving gingival manipulation, manipulation of the periapical tissue and perforation of oral mucosa
Previous IE	Respiratory tract procedures WITH transection of respiratory mucosa (e.g. tonsillectomy, adenoidectomy)
Congenital heart disease - Cyanotic CHD, unrepaired - CHD repair with patch/prosthetic within 6 months - CHD with residual defect near patch/prosthetic	Any piercing of infected skin (e.g. biopsy of rash) * NOT FOR GI/GU/Gyne procedures!
Cardiac transplantation recipients who develop cardiac valvulopathy	Regimen (give 30-60 min before dental or respiratory procedures procedure): - Amoxicillin 2g PO x1, NPO → Ampicillin 2g IV/IM OR cefazolin/ceftriaxone 1 g IV/IM - PCN allergy → Cephalexin 2g PO OR azithro 500mg PO OR doxycycline 100mg PO - NPO + PCN allergy → Cefazolin/ceftriaxone 1g IV/IM - For skin and respiratory tract procedures, target the organism which may be causing specific infection

4. Pulmonary Infections

References:

- Metlay JP, Waterer GW, Long AC, *et al.* Diagnosis and Treatment of Adults with Community-acquired pneumonia. An Official Clinical Practice Guideline of the American Thoracic Society and Infectious Diseases Society of America. *American Journal of Respiratory and Critical Care Medicine*. 200(7):**2019**;e45-67.
- Kalil AC *et al.* Management of Adults With Hospital-acquired and Ventilator-associated Pneumonia: 2016 Clinical Practice Guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clin Infect Dis*. **2016** Sep 1;63(5):e61-e111
- Uranga A, Espana PP, Bilbao A, *et al.* Duration of Antibiotic Treatment in Community Acquired Pneumonia: A Multicenter Randomized Clinical Trial. *JAMA* 2016;176(9)1257-65.
- Uyeki TM, Bernstein HH, Bradley JS, *et al.* Clinical Practice Guidelines by the Infectious Diseases Society of America: 2018 Update on Diagnosis, Treatment, Chemoprophylaxis, and Institutional Outbreak Management of Seasonal Influenza. *Clin Infect Dis*. 2019;68(6):e1-e47. doi:10.1093/cid/ciy866
- Aoki FY, Allen UD, Mubareka S, Papenburg J, Stiver HG, Evans GA. Use of antiviral drugs for seasonal influenza: Foundation document for practitioners-Update 2019. *J Assoc Med Microbiol Infect Dis Can*. 2019;4(2):60-82. Published 2019 Jun 17. doi:10.3138/jammi.2019.02.08
- Chaudhuri D, Nei AM, Rochwerg B, *et al.* 2024 focused update: guidelines on use of corticosteroids in sepsis, acute respiratory distress syndrome, and community-acquired pneumonia. *Crit Care Med*. Online special article. January 19, 2024. doi: 10.1097/CCM.0000000000006172.



Community Acquired Pneumonia (CAP)

IDSA/ATS 2019 +
2007 CAP
Guidelines Def'n
Severe CAP →

Table 1. 2007 Infectious Diseases Society of America/American Thoracic Society Criteria for Defining Severe Community-acquired Pneumonia

Diagnosis:

- Clinical (fever, productive cough, SOB, signs of consolidation) + Radiographic (infiltrate on XR)

Who to hospitalize? Use Clinical Judgement + Assess Severity of Illness:

- PSI/PORT Score (*preferred in IDSA/ATS 2019)
- CRB-65
 - Confusion
 - RR ≥ 30 per min
 - sBP < 90 or dBP ≤ 60
 - Age ≥ 65
 - Score 0-1 – Outpatient
 - Score ≥ 2 – Likely needs admission

Validated definition includes either one major criterion or three or more minor criteria

Minor criteria

Respiratory rate ≥ 30 breaths/min
PaO₂/F_IO₂ ratio ≤ 250
Multilobar infiltrates
Confusion/disorientation
Uremia (blood urea nitrogen level ≥ 20 mg/dl)
Leukopenia* (white blood cell count $< 4,000$ cells/ μ l)
Thrombocytopenia (platelet count $< 100,000$ / μ l)
Hypothermia (core temperature $< 36^{\circ}\text{C}$)
Hypotension requiring aggressive fluid resuscitation

Major criteria

Septic shock with need for vasopressors
Respiratory failure requiring mechanical ventilation

*Due to infection alone (i.e., not chemotherapy induced).

CAP – Pathogens

- Common pathogens:
 - *S. pneumoniae* (**most common**)
 - *M. pneumoniae*
 - *C. pneumoniae*
 - *H. influenzae*
 - Respiratory viruses (Influenza, RSV, parainfluenza, rhinovirus, adenovirus, coronaviruses...)
- Severe disease
 - *Legionella pneumophila*
- In patients with increasing comorbidities, antibiotic and hospital exposures:
 - Increasing gram negatives e.g. *K. pneumoniae*, *Pseudomonas* (PsA)
 - *S. aureus* (including MRSA)
 - post-influenza pneumonia
- Travel (Always ask in respiratory syndromes!)
 - Influenza, COVID-19 most common
 - *Legionella* (including non-pneumophila species)
 - MERS and other emerging viruses – need to be aware of outbreaks
 - TB, Coccidiomycosis, Histoplasmosis if right epi
 - Less common: Hantavirus, Q fever, parasitic infection

CAP – Diagnostics

IDSA/ATS CAP 2019 Guidelines:

- *Do not* obtain sputum or blood C&S on outpatients (yield low)
 - Consider for inpatients if severe CAP / intubated/ being treated empirically for MRSA or Pseudomonas
- Consider urine pneumococcal + legionella Ag +/- lower tract Legionella NAAT in severe CAP or when indicated by epidemiological factors (e.g. outbreak)
- Send rapid influenza molecular assay (NAAT) when influenza virus is circulating in community
 - *In severe CAP or immunocompromised patients, can also send NAAT for non-influenza viruses*
- *** Not included in 2019 guidelines, but current standard of care would include COVID-19 PCR*
- *Do not* send procalcitonin levels to distinguish between viral and bacterial pathogens

CAP – Outpatient Treatment



- Healthy outpatients without comorbidities or risk factors
 - **Amoxicillin 1 g TID** (strong, moderate)
 - Doxycycline 100 mg BID (conditional, low)
- Outpatients with comorbidities (chronic heart, lung, liver, renal, diabetes, alcoholism, malignancy, or asplenia)
 - Amox-clav **PLUS** macrolide (strong, moderate) OR Doxy (conditional, low)
 - If β -Lactam Allergy - Resp FQ ie. levofloxacin/moxifloxacin (strong, moderate)
 - Adds additional coverage for *H. flu* and *M. catarrhalis* (both produce beta-lactase frequently) and as well provides coverage for *S. aureus* and gram negatives, which are particularly high risk in COPD

CAP – Inpatient Treatment

- **Inpatients, non-severe, without risk factors for MRSA or PsA**

- Beta-lactam (CTX, amp-sulbactam, cefotaxime, or ceftaroline) **PLUS** Macrolide (strong, high).
 - Beta-lactam + Doxy is a third line option as alternative if unable to macrolide or FQ (conditional, low), but this is not first line because legionella has higher rates of resistance to doxycycline than Macrolide or FQ
- If β -Lactam Allergy - Resp FQ (levofloxacin, moxifloxacin)



- **Inpatients, severe CAP, without risk factors for MRSA or PsA**

- Beta-lactam **PLUS** Macrolide (strong, moderate)
- Beta-lactam **PLUS** Resp FQ (strong, low)
 - Evidence that macrolide containing combination had lower risk of death, and evidence that combination of beta-lactam and Resp FQ had higher mortality (but poor quality evidence/small number of observational trials)

- **Aspiration Pneumonia** – recommend AGAINST adding empiric anaerobic coverage unless empyema or abscess present (conditional, low quality)

CAP – Other

Mandell et al, 2007
Uranga et al, 2016
Siemieniuk RAC et al, 2015

Consider MRSA coverage based upon local risk factors

- Vancomycin 15 mg/kg IV q12h or linezolid 600 mg q12h
- Consider severe post-influenza pneumonia, mechanical ventilation/ICU, *significant* recent antibiotic

Consider Pseudomonas coverage based upon locally validated risk factors

- Pip-Tazo or Cefepime 2g q8h or Ceftaz 2g q8h or Aztreonam or Meropenem 1g q8h
- Usually recent mechanical ventilation OR prior isolation of organism.

Transition to PO

- Hemodynamically stable, improving, tolerated PO/absorbing from GI

Duration of Treatment

- 5 days if afebrile x 48 hrs with ≤ 1 sign of CAP clinical instability (HR>100, RR>24, SBP<90, paO₂<90%, can take PO, normal mental status)



Steroids

- New guidance for steroids if CAP is being managed in the ICU (SOCCM 2024 Guidelines, CAPE COD Trial, see critical care lecture)

HAP/VAP

Microbiology

- Core pathogens = *S. pneumoniae*, **MSSA**, *H. influenzae*, **GNB including *Pseudomonas*, *Stenotrophomonas***

Empiric treatment based on LOCAL antibiogram

Diagnosis

- Diagnosis based on clinical criteria alone
- Obtain non-invasive respiratory samples
- CPIS score and inflammatory markers
NOT recommended by IDSA
- Tracheobronchitis v. VAP

Table 2. Risk Factors for Multidrug-Resistant Pathogens

Per IDSA (USA guidelines, much higher prevalence overall!!!)

Risk factors for MDR VAP

- Prior intravenous antibiotic use within 90 d
- Septic shock at time of VAP
- ARDS preceding VAP
- Five or more days of hospitalization prior to the occurrence of VAP
- Acute renal replacement therapy prior to VAP onset

Risk factors for MDR HAP

- Prior intravenous antibiotic use within 90 d

Risk factors for MRSA VAP/HAP

- Prior intravenous antibiotic use within 90 d

Risk factors for MDR *Pseudomonas* VAP/HAP

- Prior intravenous antibiotic use within 90 d

Abbreviations: ARDS, acute respiratory distress syndrome; HAP, hospital-acquired pneumonia; MDR, multidrug resistant; MRSA, methicillin-resistant *Staphylococcus aureus*; VAP, ventilator-associated pneumonia.

HAP/VAP

Empiric Treatment: One of Column A +/- Column B +/- Column C

Column A:

**Base =
Pseudomonas +
MSSA coverage**

- PipTazo
- Cefepime
- Imipenem/Meropenem
- Levofloxacin*

Column B:

MRSA coverage

If risk factors for MRSA (>20% prevalence) or mortality

- Vancomycin
- Linezolid

Column C:

**2nd Anti-pseudomonal agent
(use different class than base)**

If >10% resistance to base drug, or if risk factor for Pseudomonas resistance

- Ceftazidime/Cefepime/PipTazo/Mero/Imipenem
- Ciprofloxacin/Levofloxacin
- Aminoglycoside (less preferred)
- Colistin (less preferred)

*Unreliable Pseudomonas coverage, never use for targeted Tx

Generally, duration of antibiotics = 7 days

Influenza

Who to Test?

- When Influenza is circulating in community:
 - Symptomatic outpatients if will influence treatment
 - All inpatients with acute respiratory illness
- When influenza is not circulating:
 - Symptomatic inpatients with epidemiologic link to influenza case
 - Consider in high-risk patients with acute respiratory symptoms



Who to Treat?

- Any patient hospitalized with influenza
- Any Outpatients with severe/progressive illness or risk factors
 - >65, pregnant or 2 weeks post-partum, immunocompromise, comorbidities
- Consider in other outpatients <2 days onset or with high-risk household members

How to treat?

- Start Neuraminidase inhibitor as soon as possible
 - E.g. Oseltamivir 75mg BID for 5 days
 - Benefit: Resolution of symptoms ~1 day sooner
 - Harms: Nausea/vomiting
 - Can treat for longer in immunocompromise or severe disease
- No steroids or immunomodulators

When to consider bacterial co-infection?

- Initial Severe disease (extensive pneumonia, respiratory failure, hypotension)
- Deterioration after initial improvement with antivirals
- Consider if not improving with 3-5 days of antivirals

Prophylaxis:

- In institutional outbreaks, daily oseltamivir for all patients (+/- unvaccinated staff) on impacted wards until 7 days from last case
- Post-exposure prophylaxis (Oseltamivir daily x 7d) can be considered in high-risk household contacts within 48hrs

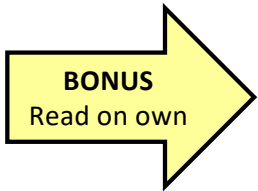
Vaccination benefits

- Influenza vaccination highly effective to reduce risk of influenza [NNT = 29 in elderly, NNT = 5 in children!] [Cochrane review 2018]
- Additionally reduces risk of hospitalization, and mortality in high risk groups who do develop influenza



Avian Influenza: H5N1

- Outbreaks of H5N1 in birds and mammals across the world in recent years, including Canada
 - Sporadic human cases, only 1 in Canada
 - No sustained human-to-human but pandemic potential
- **Primary Risk Factor is exposure to dead or ill poultry or dairy cattle**
- Incubation: 2-10 days
- Communicability: Up to 21 days
- Manifestations: Primarily respiratory, but can be GI or CNS
 - **Mortality is 52%**
- Diagnosis: NP swab, sputum or BAL for PCR with subtyping (must notify lab of suspicion!)
- Treatment: Oseltamivir (or Zanamivir)
- In hospital: Droplet/Contact Precautions, airborne for Aerosol generating procedures*
- Suspect in Patients with flu-like illness and either poultry exposure or contact of known case
- **Notify:**
 - **Lab**
 - **IPAC**
 - **Public Health**



COVID-19/SARS-CoV-2 - Diagnosis

- Main mechanism of transmission is on continuum from droplet to airborne particles
 - N95 for aerosol generating procedures
- Incubation: 2-4 days (Omicron)
- Period of Communicability: Prior to symptom onset, to around 5-10 days post onset (Omicron)
- Dx:
 - RT-PCR – URT specimen first line (70-95% SN), LRT more sensitive but more invasive
 - Rapid Antigen Test: High specificity, low sensitivity (50-85%)

Risk Factors For Severe Infection

- | | |
|----------------------|---------------------------|
| • Age >65 | • Chronic Lung Disease |
| • Pregnancy | • Cerebral Palsy |
| • Immunocompromise | • Intellectual Disability |
| • Obesity | • Sickle Cell Disease |
| • Smoking | • CKD |
| • Diabetes | • Liver Disease |
| • Heart Disease/ HTN | • Mental Health Disorders |

Complications: ARDS, myocarditis, CM, myocarditis, cytokine storm, shock, AKI, thrombosis, multisystem-inflammatory syndrome



COVID-19 Therapies in Mild Illness

Different Guidelines with different recommendations!

- To reduce risk of hospitalization; no evidence for reducing symptoms burden or duration
- Studies in **unvaccinated high-risk patients**; limited evidence in other patients
- Generally recommended in high-risk patients but definition varies between guidelines
 - Severe immunocompromise (Active cancer treatment, HSCT or CAR-T, primary immunodeficiency, advanced HIV, high dose steroids, transplant, TNF-inhibitor or immunosuppressive biologics)
 - Unvaccinated Pregnant Patients*
 - Long-Term Care (variable recommendations)
 - Other patients dependent on age, vaccine status and comorbidities (Obesity, Diabetes, heart disease, lung disease, intellectual disability, sickle cell, kidney disease, liver disease)

[Infectious Diseases Society of America Guidelines on the Treatment and Management of Patients with COVID-19](#)

[COVID-19 Science Table \(April 2022\) Clinical Practice Guideline Summary: Recommended Drugs and Biologics in Adults Patients with COVID-19](#)

[BC Clinical Practice Guide for the Use of Therapeutics in Mild-Moderate COVID-19](#)

*remdesivir in pregnancy – if immunocompromise or unvaccinated, caution re lack of data on safety

*Molnupiravir contraindicated in pregnancy (animal studies showing harm)

Treatment Options

Nirmatrelvir/Ritonavir

300/100mg BID x 5 days

*Start within 5 days

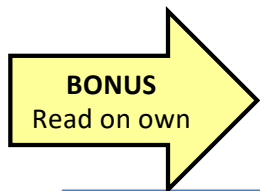
*\$1000-\$2000

Adjust for CrCl, avoid if advanced liver disease, warning re drug-drug interactions (amio, DOACs...)

Remdesivir 200mg IV x 1 day followed by 100mg daily x 2 days

*Start within 7 days

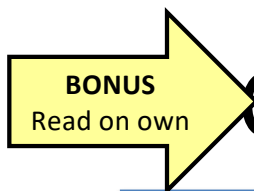
Variably Recommended Therapies: Budesonide, Fluvoxamine, Molnupiravir, Convalescent Plasma



COVID-19: Therapies

MODERATE (Low Flow Oxygen)	SEVERE (IMV, ECMO)
Dexamethasone 6mg PO daily x 10 days	Dexamethasone 6mg PO daily x 10 days
Remdesivir 200mg x 1 then 100mg daily x 4	Immunomodulators (one of): • Tocilizumab (IL-6 receptor antagonist) • Sarilumab (IL-6 receptor antagonist) • Baricitinib (JAKinhibitor)
Immunomodulators if not improving (one of): • Tocilizumab • Sarilumab • Baricitinib • Tofacitinib	

NOT RECOMMENDED: Antibiotics, Colchicine, IFN, Vitamin D, Hydroxychloroquine, Ivermectin, Lopinavir/ritonavir, Casirivimab+imdevimab or sotrovimab, Ivermectin, Famotidine, Anakinra



COVID-19 Vaccines (NACI Guidance)

	Type of vaccine	Age	(Immuno-competent)	(Immuno-compromised)	Side effects
Moderna Spikevax (LP.8.1 mRNA vaccine)	mRNA	6m+	1-2 doses (by age) 8 weeks apart	2-3 doses (by age) 4-8 weeks apart	Local site reactions and systemic S/E
Pfizer-BioNTech Comirnaty (LP.8.1)	mRNA	6m+	1-3 doses (by age) 8 weeks apart	4 doses 4-8 weeks apart	<u>myocarditis/pericarditis, Bell's palsy, anaphylaxis</u>
Novavax Nuvaxovid (JN.1 spike protein)	Protein subunit	12y+	1 dose	2 doses 4-8 weeks apart	Local site reactions and systemic S/E

- Unvaccinated Individuals 6 months to 5 years should complete series only with Moderna Spikevax
- For vaccinated individuals: Booster dose if >6 months from last vaccine dose or known COVID infection
 - Particularly: Age >65, LTC or congregate living, high risk comorbidities, pregnancy, Indigenous, Racialized communities, providers of essential community services

Contraindications to COVID-19 vaccine: Severe allergic reaction (anaphylaxis) or immediate allergic reaction to any ingredient (ie polyethylene glycol), severe or immediate allergic reaction after first dose.

[Canadian Immunization Guide](#)

5. Infectious Diarrhea

References:

- Shane AL, Mody RK, Crump JA, *et al*. Infectious Disease Society of America Clinical Practice Guidelines for the Diagnosis and Management of Infectious Diarrhea. *Clin Infect Dis* 2017;65(12):e45-80.
- Loo VG, Davis I, Embil J, et al. Association of medical microbiology and infectious disease Canada treatment practice guidelines for Clostridium difficile infection. *JAMMI* 2018. 3.2, 71-92.
-  Johnson et al IDSA & SHEA 2021 Focused Update Guidelines on Management of Clostridioides difficile infection in adults. *CID*, Volume 73, Issue 5, 1 September 2021, Pages e1029-e1044 , <https://doi.org/10.1093/cid/ciab549>
-  Kelly *et al*. ACG Clinical Guidelines: Prevention, Diagnosis and Treatment of Clostridioides difficile Infections *Am J Gastroenterol* 2021;116;1124-1147.
<https://doi.org/10/14309/ajg.0000000000001278>

Diarrhea: Diagnosis

- Stool cultures for *Salmonella*, *Shigella*, *Campylobacter*, *Yersinia*, STEC in patients with diarrhea AND:
 - Fever
 - Bloody or mucoid stools
 - Severe abdominal pain
 - Sepsis
 - Immunocompromise or outbreak exposure
 - (*V. Cholerae* in large volume rice water stools)
- *C. difficile* testing in patients with:
 - Recent antibiotics
 - Work in healthcare/LTC or prison
 - Compatible syndrome
 - IBD flare
- Blood cultures in patients with:
 - Immunocompromise
 - Sepsis
 - Suspicion of enteric fever
- Stool for Ova and Parasites in patients with:
 - Diarrhea ≥ 14 days
 - Immunocompromise e.g. HIV
 - Travel
 - *Increased yield if ordered daily x 3 days*
 - Repeat up to 3x to increase yield if high suspicion

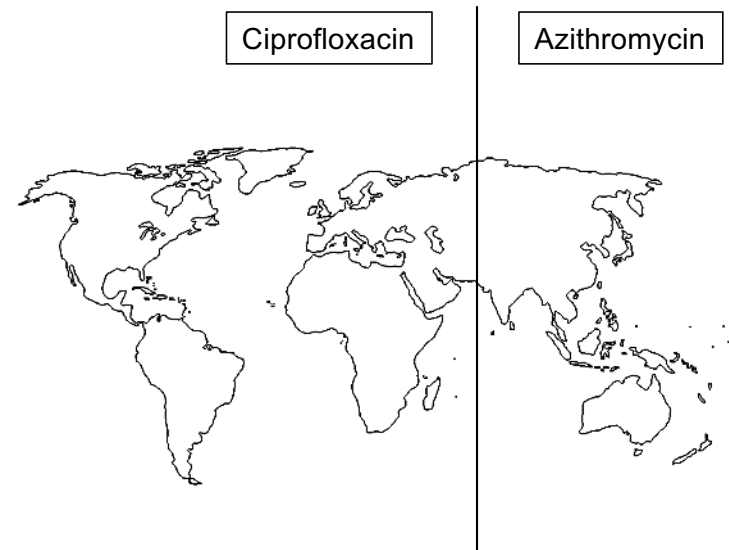
Diarrhea - Treatment

Empiric therapy in adults with bloody diarrhea not recommended UNLESS:

1. Sick immunocompetent patients with bacillary dysentery (frequent scant bloody stools, abdominal pain, tenesmus, fevers), suggestive of *Shigella*
2. Recent travel with high fever (≥ 38.5) and/or sepsis
3. Sick immunocompromised patients

Empiric antibiotic choice: ciprofloxacin or azithromycin

May use loperamide; caution if bloody stool, fever



Diarrhea – Treatment

- Modified treatment for specific pathogens (while awaiting sensitivities):

Indication	First Choice	Alternative
<i>Campylobacter</i>	Azithromycin	Ciprofloxacin
<i>S. enterica typhi</i> or <i>paratyphi</i>	Ceftriaxone OR ciprofloxacin	Ampicillin OR TMP-SMX OR azithromycin
<i>Shigella</i>	Azithromycin OR ciprofloxacin OR ceftriaxone	TMP-SMX OR ampicillin
<i>Vibrio cholerae</i>	Doxycycline	Ciprofloxacin, azithromycin, or ceftriaxone
<i>Yersina enterocolitica</i>	TMP-SMX	Cefotaxime or ciprofloxacin
Non-typhoidal <i>S. enterica</i> , STEC (inc. O157)	Not indicated	N/A

C. difficile Infection (CDI) Diagnosis

- **Testing:**

- Stool toxin test - combinations
 - EIA for GDH, toxin
 - NAAT PCR for toxin
- Pseudomembranes on colonoscopy

- **Clinical Syndrome:**

- Unexplained new-onset ≥ 3 unformed stools in 24 hours

- ***C. difficile* colonization:** Positive *C. diff* stool test in absence of clinical syndrome

- **Criteria for severe *C. difficile*:**

- WBC ≥ 15 **OR** serum Cr 1.5 x pre-morbid level
- Other risk factors: Age > 65, immunosuppression, T > 38, Albumin < 30

- **Fulminant *C. difficile*:**

- Sepsis, Shock, Ileus, perforation, toxic megacolon (colon dilation >6cm)

💡 *C. difficile* Infection First Episode Treatment

- Determine colonization vs infection (based on clinical picture)
- STOP non-essential antibiotics! Stop PPI if not needed! No evidence for probiotics.
- Fidaxomicin similar response but less recurrence after 1st episode, but expensive (not often covered)

Disease	Therapy	Recommendations
1st episode (non-fulminant)	Fidaxomicin 200 mg PO BID x 10d	First line: <i>IDSA, ACG</i> Alternative: <i>AMMI</i>
	Vancomycin 125mg PO QID x 10d (10-14d for AMMI)	First line: <i>ACG, AMMI</i> Alternative: <i>IDSA</i>
	Metronidazole 500mg PO TID x 10-14d (non-severe only)	First line: <i>ACG in low-risk pts</i> Alternative: <i>IDSA, AMMI</i>
1st episode (Fulminant)	Vancomycin 500 mg PO/NG QID + IV metronidazole 500 mg Q8H	<i>IDSA, ACG, AMMI</i>
	+/- PR vancomycin	For paralytic ileus or bowel in discontinuity
	+/- Total colectomy	



C. difficile Infection Recurrence Treatment

Disease	Therapy	Notes
First relapse (Within 3 months of previous infection)	Fidaxomicin 200mg PO BID x 10d OR BID x 5d then q2d x 20d	First Line: <i>IDSA, ACG, AMMI (if severe)</i> Alternative: <i>AMMI (if mild)</i>
	Vancomycin Taper + Pulse	First Line: <i>ACG</i> Alternative: <i>IDSA</i>
	Vancomycin 125mg PO QID x 10d (14d)	First Line: <i>AMMI</i> Alternative: <i>IDSA</i>
	Metronidazole 500mg PO TID x 10-14d	Alternative: <i>AMMI (if non-severe)</i>
	Bezlotoxumab 10mg/kg IV x 1	Adjunctive: <i>IDSA, ACG (if risk factors)</i>
≥2nd relapse	Fidax 200mg PO BID x 10d OR BID x 5d then q2d x 20d	First Line: <i>IDSA</i>
Risk factors for recurrence • Recurrent CDI in last 6m • age>65 • Immunocompromised • severe CDI on presentation	Vancomycin Taper + Pulse	First Line: <i>IDSA, AMMI</i>
	Vanco 125mg PO QID x 10d then Rifaximin 400mg x 20d	First Line: <i>IDSA</i>
	Bezlotoxumab 10mg/kg IV x 1	Adjunctive: <i>IDSA, ACG</i>
	FMT (≥3 episodes)	<i>IDSA, ACG, AMMI</i>
	Oral Vancomycin Suppression	<i>ACG (if high risk and ongoing abx; unable to have FMT, relapse after FMT)</i>

6. Intraabdominal Infections (IAIs)

- Adequate source control is #1 principle of management
 - Percutaneous if possible, laparotomy otherwise.
 - Collections 3cm or smaller can be attempted to be managed with antibiotics alone
- Initial antimicrobial coverage
 - Community-acquired, no previous hospitalization (*E. coli*, *B. fragilis*) -> Ceftriaxone or ciprofloxacin PLUS metronidazole (OR Amox-Clav)
 - Healthcare-associated or critically ill (*Pseudomonas* coverage) -> Piptazo, meropenem, ceftazidime OR cipro PLUS metronidazole
 - Enterococcal coverage for healthcare-associated or severe biliary infection, immunocompromise, post-operative infection or intravascular prosthesis/valvular heart disease. (Amox-Clav, PipTazo, Imipenem, Vanco)
 - Targeted antifungal coverage recommended for severe or nosocomial IAI if *Candida* isolated from intraabdominal or blood cultures. Empiric coverage does not improve mortality.
- **STOP-IT Trial (2015):** If source control achieved, 4 days of antibiotics has similar outcomes to continuing until 2 days after resolution of fever, leukocytosis, and ileus.
- **IDSA 2024 IAI guideline:** Addresses preferred imaging modalities and cultures for diagnosis but does not direct specifics regarding management



7. Genitourinary Infections

References:

- Gupta, K *et al.* International Clinical Practice Guidelines for the Treatment of Acute Uncomplicated Cystitis and Pyelonephritis in Women: A 2010 Update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. *Clin Infect Dis.* (2011) 52 (5):e103-e120.doi: 10.1093/cid/ciq257
- Nicolle, L *et al.* Clinical Practice Guideline for the Management of Asymptomatic Bacteriuria: 2019 Update by the IDSA. *Clin Infect Dis.* (2019) 69(10):e83-110
- Blondel-Hill, E *et al.* AMMI Canada position statement on asymptomatic bacteriuria. AMMI Canada Position statement. *JAMMI* (2018) 3(1) doi:10.3138/jammi.3.1.02
- Trautner et al. Complicated Urinary Tract Infections (cUTI): Clinical Guidelines for Treatment and Management. *Infectious Diseases Society of America* (July 2025)



BONUS
Read on own

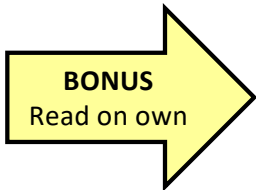
UTI – Diagnosis

- **Clinical diagnosis**, supported by urinalysis, culture +/- imaging
 - Cystitis Symptoms: Dysuria, suprapubic pain, frequency, hematuria
 - Pyelonephritis Symptoms: Fever, chills, flank pain, nausea/vomiting
 - Cloudy, smelling urine is NOT a symptom of UTI
 - Delirium without other symptoms is not considered symptomatic UTI
-  **Uncomplicated** – Infection confined to the bladder in men or women
- **Complicated** – Pyelonephritis, febrile or bacteremic, CAUTI
- Do NOT treat asymptomatic bacteruria UNLESS:
 - Pregnant
 - Urologic procedure with mucosal transection

**2025 Update**

UTI – Empiric Treatment

Syndrome	First Line Treatment	Alternatives
Acute Simple Cystitis	Nitrofurantoin 100mg BID x 5d TMP-SMX 1 DS BID x 3 days Fosfomycin 3g PO X 1	Ciprofloxacin 250mg BID x 3 days Levofloxacin 250mg daily x 3 days Amox-Clav 500/125mg BID x 5-7 days Cephalexin 250-500mg QID x 5-7 days
Complicated UTI without sepsis (Oral)	Ciprofloxacin 500mg x 7 days TMP-SMX 1 DS BID x 7-14 days	Amox-Clav 875/125 mg BID x 10-14 days
Complicated UTI without sepsis (IV)	3 rd or 4 th gen cephalosporin Piperacillin-tazobactam Fluroquinolones	Carbapenem Novel beta-lactam/beta-lactamase inhibitors Cefiderocol, Plazomicin, or Aminoglycosides
Complicated UTI with sepsis (IV)	3 rd or 4 th gen cephalosporin Carbapenem Piperacillin-tazobactam Fluroquinolones	Novel beta-lactam/beta-lactamase inhibitors Cefiderocol, Plazomicin, or Aminoglycosides



Prostatitis

- *E. coli*, Other *Enterobacteriaceae*, *Pseudomonas*
- Occasionally *Enterococci*, *S. aureus*
- Do not treat if asymptomatic unless elevated PSA, planning for biopsy or infertility
- Acute: fever, dysuria, pelvic pain, tender and edematous prostate
 - Obtain urinalysis + culture prior to Abx
 - Abx – empiric pip/tazo, 3rd gen ceph, FQ
 - Duration 2-4 weeks
- Chronic, bacterial: Subtle. Recurrent UTIs, obstructive symptoms.
 - UCx with prostatic massage
 - Abx – FQ or TMP/SMX based on susceptibility
 - Duration 4-6 weeks if FQ, 8-12 weeks if other Abx

Can Urol Assoc J
2011;5(5):306-15; DOI:10.5489/cuaj.11211

8. Sexually Transmitted Infections (STI)

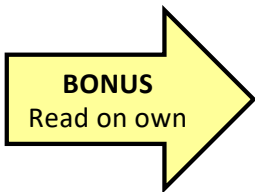
References:

- CDC (July 23, 2021) Sexually Transmitted Infections Treatment Guidelines, 2021. MMWR Vol. 70 No. 4 <https://www.cdc.gov/std/treatment-guidelines/default.htm>
-  Public Health Agency of Canada. Sexually transmitted and blood-borne infections. PHAC; Ottawa (ON): PHAC; [updated Dec 2024]. <https://www.canada.ca/en/public-health/services/infectious-diseases/sexual-health-sexually-transmitted-infections/canadian-guidelines.html>

MCQ 1 2026

A 23 year old man presents with a new diagnosis of urethral gonorrhea, and testing for Chlamydia was not performed. What is the appropriate initial treatment according to PHAC guidelines:

- a) Ceftriaxone 500 mg IM x1 dose plus Azithromycin 1 g PO x1 dose with test of cure
- b) Ceftriaxone 250 mg IM x1 dose plus Azithromycin 1 g PO x1 dose with test of cure
- c) Ceftriaxone 250 mg IM x1 dose plus Azithromycin 1 g PO x1 dose without test of cure
- d) Ceftriaxone 500 mg IM x1 dose with test of cure



Gonorrhea/Chlamydia

	History	Investigations	Complications
<i>Neisseria gonorrhea (NG)</i>	Incubation 2-7 days. Rectal/pharynx infxn more likely to be asymptomatic. Men: usually symptomatic Urethral discharge, dysuria, urethral itch, testicular pain/epididymitis, rectal pain/proctitis Women: Often asymptomatic Vaginal/cervical discharge, low abdo pain, dysuria, dyspareunia, rectal pain/proctitis	NAAT/Culture from appropriate anatomical site - Urine - Cervical swab - Rectal Swab - Pharyngeal swab For DGI: - Screen ALL exposed sites (above) AND all disseminated sites - E.g. Synovial Fluid/Blood culture +/- pustule swab *Culture preferred for resistance testing.	M: Urethral strictures with fistulas, epididymitis (Chlamydia > gonorrhea), infertility (rare) F: PID, infertility, ectopic pregnancy, Fitz-Hugh-Curtis syndrome Both: Re-infection, reactive arthritis, DGI = arthritis, dermatitis, endocarditis, meningitis (more common in pregnant women or MSM), increased risk of HIV
<i>Chlamydia trachomatis (CT)</i>	Incubation ~2-6 weeks after infection. May persist for months if untreated 50% of Men and 70% of women are asymptomatic Genital infection symptoms similar to gonorrhea above LGV More invasive	NAAT from appropriate specimens - Urine - Cervical swab - Rectal Swab - Pharyngeal swab Send specimen for genotyping if concern for LGV (e.g. rectal infection)	

Gonorrhea/Chlamydia Treatment

Anogenital or Pharyngeal:
PHAC 2024 (now in agreement with CDC 2021)

Other:

Gonorrhea



Ceftriaxone 500mg IM x 1
 (Alt: Cefixime 800mg PO x 1)

PLUS

Azithromycin 1g PO x 1
 OR
 Doxycycline 100mg PO BID x 7d

For complicated infections (PID, epididymitis, DGI, ophthalmia):

Ceftriaxone 2g IM/IV q24h (duration depends on syndrome)

PLUS

Azithromycin 1 g PO x1 dose

TOC of cure after treatment for all infections (culture if <3 days, NAAT if >3-4 weeks after)

Chlamydia

Azithromycin 1g PO x 1
 OR
 Doxycycline 100mg PO x 7 days

For LGV:

Doxycycline 100mg PO BID x 21 days

Indications for TOC (3w after treatment) :

- LGV
- Unclear compliance
- Persistent symptoms
- Pregnant patient
- Prepubertal patient

Notes:

- 1) Doxycycline contraindicated in pregnancy (azithro is OK)
- 2) Test/Treat partner!
- 3) Counsel patient to abstain from sex x 7d AND partner treated

Syphilis

Stage	Manifestations	Treatment
Primary (3 weeks, up to 90d)	Painless chancre, regional LN	<u>PRIMARY, SECONDARY, EARLY LATENT:</u> Benzathine penicillin G 2.4 mU IM x 1 <i>ALTERNATIVE (limited evidence):</i> · Doxycycline 100mg BID x 14d · Ceftriaxone 1g IM/IV x 10d
Secondary (12w - 6 months)	Fever, malaise, rash, alopecia, uveitis, meningitis, LN, hepatitis, arthralgias, condylomata lata	
Latent Early (< 1 year)	Positive serology	<u>LATE LATENT or UNKNOWN DURATION, TERTIARY SYPHILIS:</u> Benzathine penicillin G 2.4 mU IM weekly x3 <i>ALTERNATIVE:</i> · Doxycycline 100mg BID x 28d · Ceftriaxone 1g IM/IV x 10d *if carefully monitored*
Late (> 1 year), or unknown duration	No clinical manifestations	
Tertiary	- Cardiovascular - Aortitis - Gummatous - Late neurosyphilis - Tabes dorsalis - General paresis	<u>NEUROSYPHILIS</u> Aqueous penicillin 4mU q4 hours IV x 14 days <i>ALTERNATIVE:</i> · Preferably desensitize · Ceftriaxone 1-2g IM/IV x 10-14d *With specialist*

Neurosyphilis can occur at any stage!

Asymptomatic, meningitis, otic symptoms, headaches, ocular symptoms, Argyl-Robertson pupil, dementia...

For PCN allergy, consider desensitization for:

****Pregnancy**, Neurosyphilis**

Syphilis Testing

- TT = Treponemal test (immunoassays, TPPA, FTA-ABS)
 - Specific antibody against *T. pallidum*, persist over lifetime
- NTT = Non-treponemal tests (VDRL, RPR)
 - Non-specific antibody released during infection

Indications for LP:

- Neurologic, ocular, or auditory symptoms
- Inadequate serologic response to treatment
- Tertiary syphilis
- (HIV with CD4 < 350 or RPR >1:32 maybe)

Who to test (PHAC, 2023):

- Sexually active adults (q3-6months)
- Pregnant people (first visit, again at 28-32w, and once again at time of delivery only if in an outbreak area if at risk of reinfection)
- Suggest q3month opt-out programs for high-prevalence communities such as:
 - Gay, bisexual, and other MSM
 - People living with HIV
 - Person who is or was incarcerated
 - People who use substances or access addiction services
 - Some Indigenous communities

CMIA (TT)	RPR (NTT)	TPPA (TT)	Interpretation
-	Not Done	Not Done	Negative or early incubating syphilis
+	-	-	False positive
+	+	-	early infection, late latent/tertiary, previously treated
+	-	Indeter	Inconclusive, repeat
+	+	Indeter	Inconclusive, repeat
+	-	+	Previously treated , early infection, late latent/tertiary
+	+	+	Current or previously treated infection

MPOX

- 2 clades – Clade I (a or b) and Clade II (a or b)
 - Clade IIb implicated in 2022 outbreak, more severe Clade Ia and Ib in Africa since 2024
- Transmission (humans or infected animals):
 - **Direct Contact with infectious sores or body fluids (often during sexual or intimate contact)**
 - Fomite Transmission, e.g. linens
 - Respiratory secretions
 - Vertical Transmission
- Droplet/Airborne Precautions
- Diagnosis: PCR (swab of lesion/fluid)
- Most Self-Resolve
 - Off-label Tecovirimat (TPOXX) if complications or high risk
- Vaccines:
 - Partial protection if prior smallpox vaccine
 - Imvamune for high risk individuals or as post-exposure prophylaxis in high risk exposures

Incubation: 5-21 days

Prodrome Phase: 0-5 days

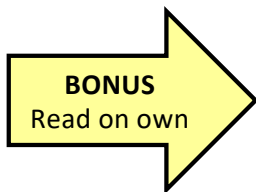
- | | |
|-------------------|---------------|
| • Fever | • Arthralgias |
| • Chills | • Back Pain |
| • Lymphadenopathy | • Weakness |
| • Headache | • Fatigue |
| • Myalgias | |

Skin Eruption Phase:

1-3 days post fever onset, lasts 2-4 weeks

- Face, extremities but can affect hands, feet, mouth, genitals and perianal
- Progression: macules -> papules -> vesicles -> pustules -> ulcers -> scab
- Lesions tend to evolve synchronously
- Painful lesions (may be pruritic during healing)

Infectious: Symptom onset until scabs fall off



Extra STIs

	Notes	Treatment
M. genitalium	Persistent or recurrent urethritis/cervicitis with negative G/C NAAT	Azithromycin 500mg PO x 1 then 250mg daily on day 2-5 Moxifloxacin 400mg PO daily x 7 days
Chancroid (H. ducreyi)	Painful ulcer with granulomatous base that bleeds, painful inguinal LN	Azithromycin 1g PO X 1, CTX 250mg IM x1, or Ciprofloxacin 500mg PO BID x 3d
LGV (Chlamydia)	Serovars (L1-L3) are more invasive; painful LN, hemorrhagic proctitis	Doxycycline 100mg PO BID X 21d; Rx partner(s)!
Genital HSV	HSV-1/2, painful vesicles/ulcers with prodrome (itch/tingling/burning)	Acyclovir, valacyclovir, famciclovir
Anogenital warts (HPV)	Strains 6,11; asymmetric, polymorphic lesions	Spontaneous resolution; consider imiquimod/cryotherapy. VACCINE!
Trichomonas vaginalis	Vaginal pH > 4.5; neg whiff; yellow, frothy discharge; "strawberry cervix"	Metronidazole 2g PO X 1 dose (M) OR 500mg PO X 7 days (F); Rx partner(s)!
Vulvovaginal candidiasis	Vaginal pH < 4.5; neg whiff; wet mount with 10% KOH – budding yeast	Clotrimazole, miconazole cream OR Fluconazole 150mg PO X 1 dose
Bacterial vaginosis	Vaginal pH > 4.5; pos whiff; clue cells on gram stain; fishy odor	Metronidazole 500mg PO BID X 7days OR 0.75% (5g) application PV x 5 days

MCQ 1 2026

A 23 year old man presents with a new diagnosis of urethral gonorrhea, and testing for Chlamydia was not performed. What is the appropriate initial treatment according to PHAC guidelines:

- a) **Ceftriaxone 500 mg IM x1 dose plus Azithromycin 1 g PO x1 dose with test of cure**
- b) Ceftriaxone 250 mg IM x1 dose plus Azithromycin 1 g PO x1 dose with test of cure
- c) Ceftriaxone 250 mg IM x1 dose plus Azithromycin 1 g PO x1 dose without test of cure
- d) Ceftriaxone 500 mg IM x1 dose with test of cure

This is a new dose update for PHAC treatment guidelines in the last year in order to address increasing concern for ceftriaxone resistance. Since testing for Chlamydia was not performed in this case, the recommendation is to add the single dose of Azithromycin. All cases of gonorrhea should have a test-of-cure.

9. Skin and Soft Tissue Infections

References:

- Stevens DL, Bisno AL, Chambers HF, *et al.* Practice Guidelines for the Diagnosis and Management of Skin and Soft Tissue Infections: 2014 Update by the Infectious Diseases Society of America. *Clin Infect Dis* 2014.
- Canadian Immunization Guide <http://www.phac-aspc.gc.ca/naci-ccni/>
- Public Health Agency of Canada. Guidelines for the Prevention and Control of Invasive Group A Streptococcal Disease. *CCDR* 2006;32S2:1-26.

Managing SSTIs: Purulent

- Purulent (Furuncle/Carbuncle/Abscess)
 - I&D and C&S should be performed

****Although 2014 IDSA guidelines suggested forgo antibiotics in mild abscesses with source control, multiple RCTs have since shown that a short course of antibiotics is associated with higher rates of cure and lower rates of recurrence (regardless of size, organism or severity).**

Severity	Empiric Rx	Defined Rx
Mild	<i>Consider antibiotics based on clinical context**</i>	
Moderate (systemic signs of infection)	Cephalexin* or TMP/SMX or Doxycycline	MRSA → TMP/SMX MSSA → Cephalexin
Severe (IC, failed prior antibiotics or I&D, systemic signs of infection)	Vancomycin	MRSA → Vancomycin MSSA → Cefazolin

** Do not use cephalexin if high MRSA prevalence
Stevens DL et al, 2014*

Managing SSTIs: Non-Purulent

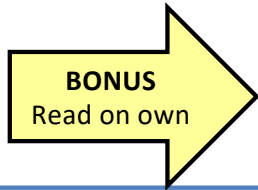
- Non-Purulent (Impetigo/Erysipelas/Cellulitis) – Think Strep!

Severity	Abx	Duration
Mild (no systemic signs of infection; no focus of purulence)	Oral cephalosporin (cephalexin)	Generally 5 days, extend up to 14 if no improvement at completion (not if simply persistent redness)
Moderate (systemic signs of infection)	IV cephalosporin (cefazolin)	

* Treat predisposing trauma, tinea pedis, xerosis, lymphedema etc.

** Consider antimicrobials effective against MRSA (and streptococci) if cellulitis is associated with penetrating trauma, evidence of MRSA infection elsewhere, nasal colonization with MRSA, PWID

*** If patient is severely immunocompromised, consider broadening coverage to Vanco + PipTazo OR Vanco + Meropenem/Imipenem



Recurrent Cellulitis Trials

PATCH I Trial (2013)

- Oral penicillin daily for at least 1 year was effective in preventing subsequent episodes of cellulitis in patients with at least 2 episodes of cellulitis in the past 3 years
- Patients with BMI > 33, ≥3 episodes per year, and lymphedema may have less effect
- **BOTTOM LINE:** Guidelines suggest considering trial of oral penicillin if ≥3 episodes/ year DESPITE controlling other risk factors (e.g. revascularization, wound care, foot wear, compression, tinea)

(Thomas et al, 2013)

Compression Stockings (2020)

- In patients with chronic leg edema and recurrent cellulitis, compression therapy was associated with reduced recurrence compared to conservative treatment
- **BOTTOM LINE:** Compression therapy for patients with chronic leg edema and recurrent cellulitis as part of first line prevention measures

(Web et al, 2020)

Necrotizing Fasciitis (NF)

Erythema with systemic toxicity, gangrene/anesthesia, hard induration, hemorrhagic bullae, pain-out-of-proportion and extending beyond erythema

- **EMERGENT** surgical inspection/debridement to rule out necrotizing process
(**SURGERY CONSULT – This is a *surgical* emergency**)
- **EMPIRIC Abx: PipTazo + Vancomycin + Clindamycin**
- Consider IVIG if shock or pre-operative

Pathogen	Defined Abx	Additional Notes
GAS (<i>S. pyogenes</i>) (TYPE 2 NF)	PCN + clindamycin	Usually younger, following minor trauma/bruise, may have DM/PAD/PWID, NSAIDs
Polymicrobial (TYPE 1 NF)	PipTazo + vancomycin or carbapenem	Usually older, DM, pelvic wounds
<i>Clostridium spp.</i>	PCN + clindamycin	Trauma-associated, Colon Ca
<i>Vibrio vulnificus</i>	Doxy + ceftazidime	Saltwater exposure ; consider if underlying liver dz, seafood ingestion
<i>Aeromonas hydrophila</i>	Doxy + ceftazidime	Freshwater exposure/injury

CANNOT CLINICALLY
DISTINGUISH TYPES!

Toxic Shock Syndrome (TSS): Group A Strep

(and sometimes *S. aureus* – tampons, nasal packing)

Diagnostic Criteria:

- Hypotension (sBP < 90) AND
- Isolation of GAS from normally sterile site

AND *at least two* of the following:

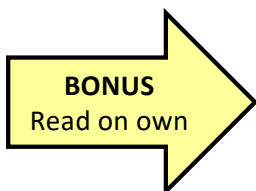
- Coagulopathy (plt < 100 or DIC)
- Liver enzymes (ALT/AST/Tbili 2x ULN)
- ARDS
- Rash - erythematous macular rash that may desquamate
- Kidney impairment (Cr > 177)

Management:

- Contact and droplet precautions*
- Volume resuscitation
- Surgical source control (especially if necrotizing SSTI suspected)
- Antibiotics: Beta lactam PLUS clindamycin
- IVIG – limited evidence but consider if severe infection
- Hyperbaric O2 (HBO)–efficacy unknown
- Chemoprophylaxis – cephalexin x10d (clinda if PCN allergy)*

*For all invasive GAS

Guidelines for prevention and control of invasive GAS disease (CCDR, 2006)



SSTI Miscellaneous Exposures/Syndromes

Clinical picture	Microbiology – classic association
Diabetes	Polymicrobial, often <i>Pseudomonas</i> and anaerobes
Water exposure	<i>Vibrio vulnificus</i> (salt water), <i>Aeromonas</i> spp. (fresh water), <i>M. marinum</i> (fish tank exposure), <i>M. fortuitum</i> , (hot tub exposure)
Rose gardens	<i>Sporothrix schenckii</i>
Meat, butchers, veterinarians	<i>Erysipelothrix rhusiopathiae</i>
Ecthyma gangrenosum; malignant/invasive otitis externa; hot tub folliculitis; green nail syndrome	<i>Pseudomonas aeruginosa</i>
“Herpetic” whitlow	HSV-1, HSV-2
Burrows, pruritic and tracks seen in web spaces	Scabies
Black eschar in nasal mucosa or palate of diabetic/ elevated glucose/immunosuppressed	<i>Mucormycosis</i> (<i>Rhizopus</i> spp.)

10. Bone & Joint Infections

References:

- Senneville É, Albalawi Z, van Asten SA, et al. IWGDF/IDSA guidelines on the diagnosis and treatment of diabetes-related foot infections (IWGDF/IDSA 2023). *Clinical Infectious Diseases*, ciad527, <https://doi.org/10.1093/cid/ciad527>
- Berbari EF, Kanj SS, Kowalski TJ, et al. 2015 Infectious Diseases Society of America (IDSA) clinical practice guidelines for the diagnosis and treatment of native vertebral osteomyelitis in adults. *Clin Infect Dis* 2015.
- Butalia S, Palda VA, Sargeant RJ, et al. Does this patient with diabetes have osteomyelitis of the lower extremity? *JAMA* 2008;299:806-13.
- American Academy of Orthopaedic Surgeons Evidence Based Clinical Practice Guideline for Diagnosis and Prevention of Periprosthetic Joint Infections <https://www.aaos.org/pjicpg> Published March 11, 2019

NOTE: Septic Arthritis is covered in the RHEUMATOLOGY LECTURE**

Osteomyelitis

Etiology

- Hematogenous (more common in children [long bones] vs. adults [vertebrae]) → monomicrobial
- Contiguous from SSTI, trauma, or surgery (more common in adults) → polymicrobial

Patient Population	Organism
All patients	<i>S. aureus</i>
Foreign body, prosthetic infection	CNST, <i>Cutibacterium acnes</i>
Nosocomial	<i>Pseudomonas</i> , <i>Enterobacterales</i> , <i>Candida</i>
Diabetes	<i>Streptococcus</i> , Gram negative bacilli, anaerobes
Immunocompromised	<i>Candida</i> , <i>Aspergillus</i> , <i>Mycobacterium</i>

Non-Vertebral Osteomyelitis (w/o hardware)

	Non-hematogenous	Hematogenous
Investigations	Blood Cultures Appropriate Imaging CRP Bone Biopsy/OR Specimen if possible	
Surgical Management	Debridement of necrotic material. Send for culture Soft tissue coverage of site of infection required for successful cure	Usually not required. Consider if subperiosteal collection, necrotic bone, concomitant joint infection.
Empiric Therapy	Ceftriaxone +/- Vancomycin (if MRSA risk factors) +/- Metronidazole (if sacral) Tailor based on organism if possible	
Duration	6 weeks from last debridement if residual infection 48 hours post complete source control	4-6 weeks

Diabetic Foot Infection

IDSA Guideline Senneville et al, 2023

Diagnosis: Diagnose soft tissue infection clinically

- CRP +/- ESR +/- Procalcitonin if equivocal exam
- Combination of probe-to-bone, plain x-rays and CRP/ESR to diagnose osteomyelitis; send MRI if diagnosis in doubt
- If possible obtain samples from tissue specimen (soft tissue) or bone biopsy (osteomyelitis) for culture
- Stage using the IWGDF/IDSA classification (**do not memorize!**)
- Consider hospitalizing patients with severe infection or mod infection w/ comorbidities

Surgery and Debridement

- Urgent Surgical consult if severe or moderate DFI with complications, early surgery to remove infected and necrotic tissue
- Consider surgery in patients with osteomyelitis or PAD with ulcer, gangrene
- Exception: Forefoot osteomyelitis only, no PAD, no exposed bone and no need for surgery to control infection

Revascularization

- Obtain vascular consult for DFI w/ PAD for consideration of revascularization

Other Pillars of Multidisciplinary Management

- Wound Care
- Glycemic Control
- Off-loading + Foot care (Chiroprapist, shoe care)

Diabetic Foot Infection

IDSA Guideline Senneville et al, 2023

Antibiotic Management

- **Agent:** Select based on likely or proven causative pathogens, severity of infection, evidence of efficacy and risk of adverse effects

Mild DFI	No recent Abx	Aerobic GPC	(eg) Cephalexin, Septra, Doxycycline, Moxifloxacin
	Recent Abx	GPC + GNB	Amox-clav, septraxone, moxifloxacin
Mod-Severe DFI	No complications	GPC + GNB	Amox-clav, Ceftriaxone
	Ischemia/necrosis	GPC + GNB + anaerobes	Amox-clav, PipTazo, Ceftriaxone + Clindamycin
	Macerated ulcer	GPC + GNB + Pseudomonas	PipTazo, Cephalexin + Cipro

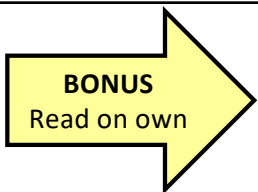
- **Duration:**
 - Soft Tissue: 1-2 wks, up to 3-4 wks if extensive/slow to resolve
 - Osteomyelitis: 48hrs (complete amputation), 3 weeks (amputation/resection with positive margins), or 6 weeks (no resection)
- **Oral Stepdown**

BONUS

Read on own

Chronic Wounds

- What type of foot ulcer is it?
 - Neuropathic: Pressure points, punched out appearance, deep ulcer, minimal pain, warm and dry foot
 - Arterial: Lateral malleolus, dry and punctate, decreased pulses, cold and dry foot
 - Venous: Medial malleolus, irregular margins, shallow depth, mildly painful, venous stasis dermatitis/lipodermatosclerosis
- Is the ulcer infected?
 - **Pain in chronic wound (LR 11-20)**
 - Foul odor (LR 1-3)
 - Purulence, exudate, erythema, warmth, and edema (LR < 1.0)



JAMA: Does this patient with diabetes have OM of the lower extremity?

Clinical Findings	LR+	LR-
Ulcer area > 2cm ²	7.2	0.48
Positive 'probe to bone'	6.4	0.39
Bone exposure	9.2	0.70
Investigations that may increase likelihood		
ESR > 70 mm/h	11	0.34
Abnormal X-ray	2.3	0.63
Role of MRI		
Positive MRI	3.8	0.14

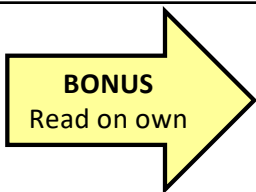
- **Gold standard = bone biopsy and culture**
- Presence or absence of ulcer inflammation (erythema, swelling, pus) does not modify probability of dz
- Superficial swab cultures do not reliably predict bone microorganisms or diagnose infection

Native Vertebral Osteomyelitis

Berbari, 2015

- **Etiology:** hematogenous seeding disc → bone
- **Microbiology:**
 - **Most common *S. aureus****
 - Beta-hemolytic streptococci, GNB
 - TB, Brucella, fungi much less common
- **Risk factors:**
 - Elderly, immunocompromised, IDU, PICC/Ports
- **Signs and symptoms:**
 - New/worsening back pain and suggestive b/w
 - Fever (only 45% of patients)
- **Diagnosis:**
 - Blood cultures (50% pos if *S. aureus*), **biopsy**
 - ESR, CRP (sensitivity 94-100%)
 - MRI (SN 97%, Sp 93%)
- **Treatment:**
 - Hold antibiotics until biopsy result if no sepsis/neuro compromise (dx 50-60% of time with 1st biopsy)
 - Empiric: ceftriaxone + vancomycin
 - Duration: 6 weeks
 - Surgery if neuro deficits, spinal cord compression, progression/recurrence despite appropriate antibiotics
- **Follow-up:**
 - Monitor clinically and repeat inflammatory markers
 - Repeat MRI ONLY if poor clinical response after ABx

*If *S. aureus* bacteremia in prior 3 months, biopsy likely not needed



PJI for the Internist

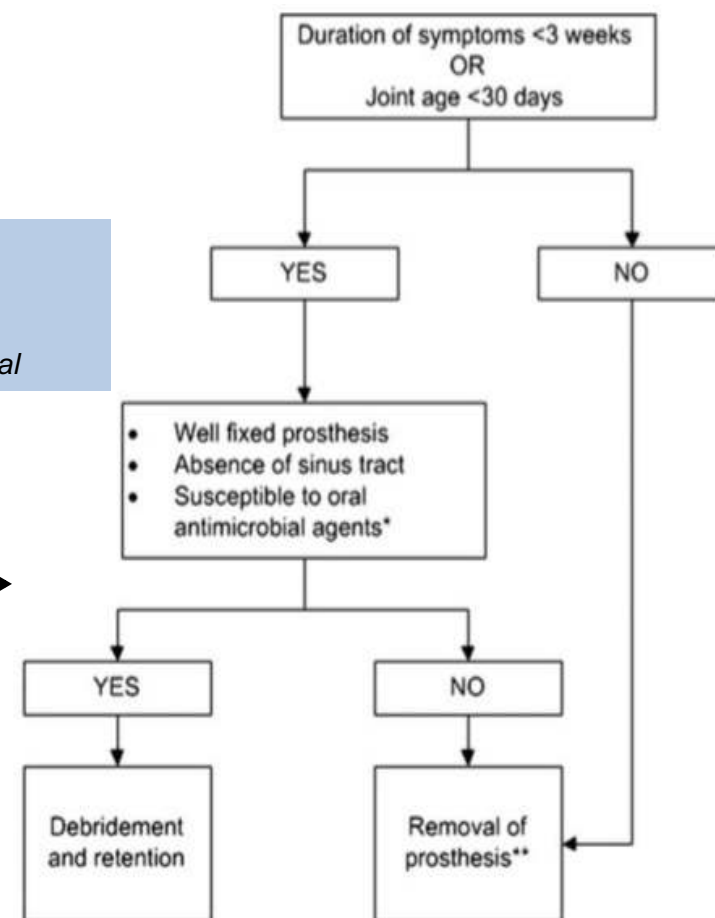
Principles:

- 1) Make micro diagnosis.
 - Withhold antibiotics if stable pending arthrocentesis / OR to ensure pathogen determined.
- 2) Surgical Management
- 3) Duration dependent on surgery + organism

	Retained Hardware	1 Stage Revision	2 Stage Revision
Staphylococcal	2-6 weeks IV followed by PO for total 6m (knee) or 3m (all else)	2-6w IV followed by PO for total 3m	4-6w IV/PO
Other	4-6 weeks IV/PO	4-6 weeks IV/PO	4-6 weeks IV/PO

Empiric Abx in a scenario:

Vanco + CTX
Tailor based on culture
Add Rifampin for Staphylococcal



Osmon et al, 2012

11. HIV

References:



- Panel on Antiretroviral Guidelines for Adults and Adolescents. Guidelines for the Use of Antiretroviral Agents in Adults and Adolescents with HIV. Department of Health and Human Services. Updated September 2025. Available at <https://clinicalinfo.hiv.gov/en/guidelines/adult-and-adolescent-arv>.
- Panel on Guidelines for the Prevention and Treatment of Opportunistic Infections in Adults and Adolescents with HIV. Guidelines for the Prevention and Treatment of Opportunistic Infections in Adults and Adolescents with HIV. National Institutes of Health, Centers for Disease Control and Prevention, HIV Medicine Association, and Infectious Diseases Society of America. 2023. Available at <https://clinicalinfo.hiv.gov/en/guidelines/adult-and-adolescent-opportunistic-infection>.
- Gandhi RT, Bedimo R, Hoy JF, et al. Antiretroviral drugs for treatment and prevention of HIV infection in adults: 2022 recommendations of the International Antiviral Society-USA Panel. *JAMA*. 2022; doi:10.1001/jama.2022.22246.



- Horberg et al. Primary Care Guidance for Providers Who Care for Persons With Human Immunodeficiency Virus: 2024 Update by HIVMA/IDSA. *CID*. 2024; <https://doi.org/10.1093/cid/ciae479>
- Wood E, Kerr T, Roswell G, et al. Does this adult patient have early HIV infection? *JAMA* 2014;312:278-85.
- Panel on Treatment of HIV During Pregnancy and Prevention of Perinatal Transmission. Recommendations for the Use of Antiretroviral Drugs During Pregnancy and Interventions to Reduce Perinatal HIV Transmission in the United States. Department of Health and Human Services. Updated June 2025. Available at <https://clinicalinfo.hiv.gov/en/guidelines/perinatal>.

MCQ 2 2026

42 year old man presents to hospital with new diagnosis of HIV and is started on ART. Which of the following would not be a part of the initial assessments in the months following the start of therapy:

- a) Starting a statin if his 10-year ASCVD risk estimates are <5%
- b) Screening for HPV-related anal cancer
- c) Vaccination for HepB if there is no evidence of prior infection or vaccination on serology
- d) Vaccination with PCV-13 followed by PCV-23 at least 8 weeks later



Approach to the New HIV Diagnosis

1. Stage the HIV Infection

- ☐ Confirm positive test
- ☐ CD4 count
- ☐ Viral load (RNA)
- ☐ Genotype (resistance)
- ☐ Tropism testing (CCR5)

2. Assess for Opportunistic Infx (OI)

- ☐ Prophylaxis / Treatment
Depending on CD4 count

3. Assess for Co-Infections

- ☐ TB: TST/IGRA, CXR
- ☐ Serologies
 - ☐ CMV IgG
 - ☐ Toxoplasma IgG
 - ☐ Hepatitis B/C serology
 - ☐ Measles serology
 - ☐ Varicella serology

☐ STI and sexual health:

- ☐ Syphilis screen
- ☐ Chlamydia, gonorrhea
NAATs +/- cultures
- ☐ HPV – Cervical and anal
cancer screening

4. Immunizations

- ☐ Hepatitis A (only at risk)
- ☐ Hepatitis B (if non-immune)
- ☐ Annual influenza + COVID
- ☐ HPV
- ☐ Pneu-C-20 or Pneu-C-21
(Hib, meningococcus)
- ☐ RSV

*Caution with live vaccines (varicella, VZV, MMR) if CD4 < 200

5. Assess general health

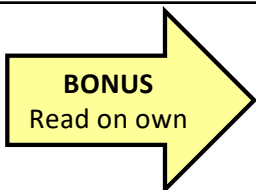
- ☐ CBC, lytes, Cr, PO4
- ☐ Non-fasting lipids/gluc
- ☐ Statin initiation - REPRIEVE
- ☐ Urinalysis/ B-HCG
- ☐ AST, ALT, ALP, Bilirubin

6. Initiate ARV

- ☐ Drug safety screening: HLA-B5701 (abacavir hypersensitivity), G6PD level
- ☐ ASAP ARV start (see slides)
- ☐ Side effects of therapy

7. Follow-up & Counselling

- ☐ Prognosis
- ☐ Safe Sex, U=U
- ☐ HIV in pregnancy
- ☐ Duty to disclose HIV status



BONUS: JAMA: Does this patient have early HIV infection?

(*check out our **2015 video** demonstrating this exam online!*)

- Primary HIV infection (Acute retroviral syndrome) → fever, nausea, emesis, weight loss, arthralgia/myalgia, pharyngitis, oral ulcers, rash and LN

*****Limited utility of clinical examination to detect or rule out early HIV infection highlights the importance of routine testing for HIV infection among adults (current recommendation is yearly screening for all adults, although practically this is not done)**

Clinical Findings	LR+	LR-
Symptoms that increase likelihood		
Genital ulcers	5.4	
Weight loss	4.7	
Vomiting	4.6	
Swollen LNs	4.6	
Fever	3.4	0.74
Physical exam signs that increase likelihood		
Presence of lymphadenopathy	3.1	0.70

Wood et al, 2014

HIV – Initiation of ARV

- Assess readiness for treatment
- **ARV recommended for all individuals with HIV, regardless of CD4 count** to reduce morbidity and mortality associated with HIV infection
- Historically, ARV regimen should include **TWO (sometimes one) NRTIs PLUS INSTI OR NNRTI OR PI**

First Line Therapy in PrEP naive:

- BIC + TAF + FTC

(eg) Biktarvy – single pill combo therapy

- DTG + (TAF or TDF) + (FTC or 3TC)
- DTG + 3TC *with caveats*

First Line Therapy with history of PrEP:

- DRV/c or DRV/r + (TAF or TDF) + (FTC or 3TC)

TAF has fewer bone and renal toxicities, whereas TDF is associated with lower lipid levels and lower cost

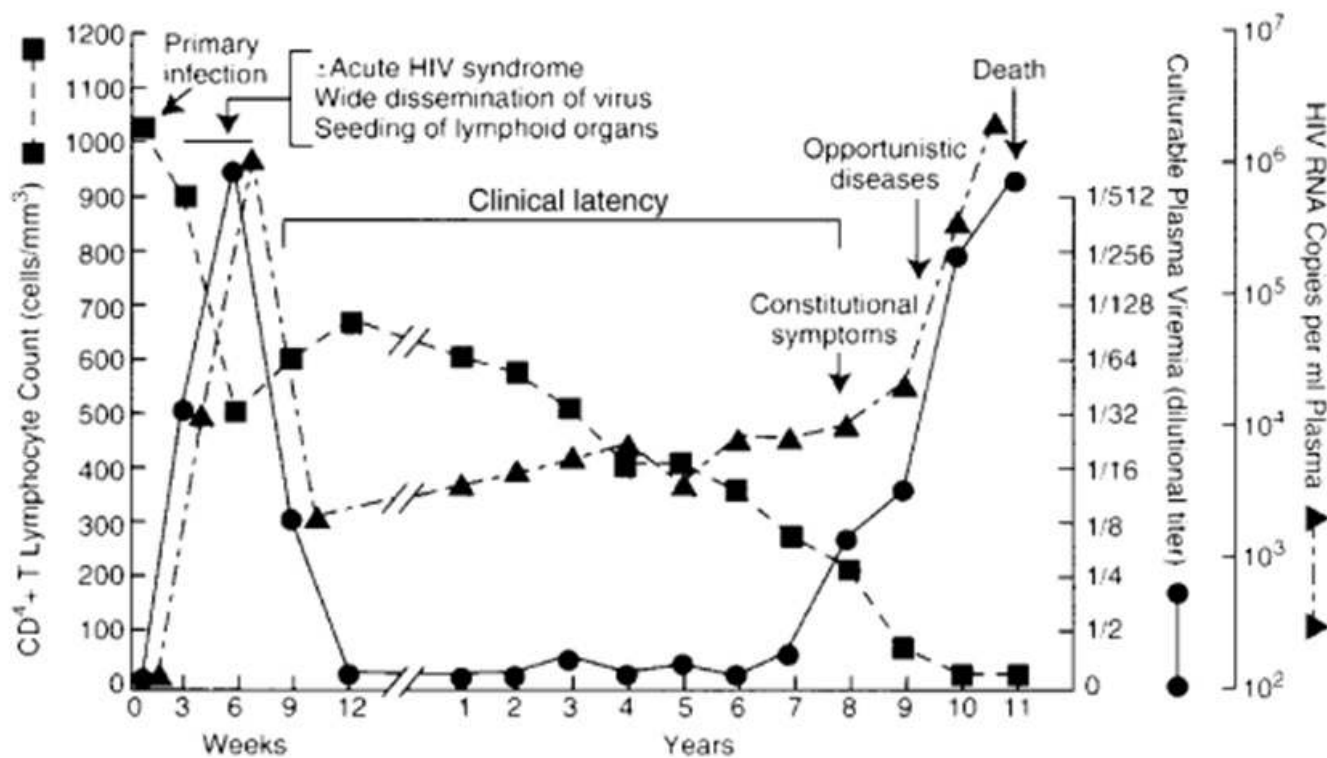
CAB + RPV long acting injectable is now seeing use in Canada, but is not included in all guidelines yet and would not be for new ARV starts

BIC = Bictegravir | TAF = Tenofovir Alafanamide | FTC = Emtricitabine
DTG = Dolutegravir | TDF = tenofovir | 3TC = lamivudine

AIDS Info Guidelines, 2025
Ghandi et al. 2020

BONUS
Read on own

HIV – Opportunistic Infections



HIV – OI Primary Prophylaxis (1)

CD4	OI	Preferred	Alternative
<200	PJP	• TMP-SMX 1 DS PO daily (AI) • TMP-SMX 1 SS PO daily (AI)	• TMP-SMX 1 DS PO M/W/F (BI) • Dapsone (check G6PD) • Atovaquone (\$\$\$) 1500mg PO daily • Aerosolized pentamidine monthly (infrastructure)

- TMP-SMX should be continued if patients have non-life-threatening reactions where feasible, or re-challenged when mild reaction resolved – consider lower dose
- Dapsone OK for sulfa allergy, NOT OK for SJS/TEN to TMP-SMX
- SJS/TEN is clear contraindication to re-challenge/continuation – give atovaquone
- **Prophylaxis with TMP-SMX recommended during pregnancy** – supplement with folic acid during first trimester (NT defects)

*** Continue prophylaxis until CD4 count stabilizes >200 for at least 3 months**

HIV – OI Primary Prophylaxis (2)

CD4	OI	Preferred	Alternative
<100	Toxoplasma (if Toxo IgG positive)	TMP/SMX 1 DS PO daily (AI)	- TMP-SMX 1 DS PO M/W/F (BIII) - TMP-SMX 1 SS PO daily (BIII) - Dapsone + pyrimethamine (+ leucovorin) (BI) - Atovaquone (CIII) (\$\$\$)
<50	MAC	- Azithromycin 1200 mg PO weekly - Clarithromycin 500mg PO BID	Rifabutin

***No longer recommended**
unless patients are not starting
ART or not on fully suppressive
ART.

Continue prophylaxis until CD4 count stabilizes for at least 3 months

HIV – OI Treatment (1)

OI	Preferred Rx	Alternative Rx
PJP	TMP-SMX 15-20mg/kg IV [TMP component] x21 days (Any severity) PLUS (FOR SEVERE ONLY): • Prednisone 40mg PO BID x5d then • 20mg PO BID x5d then • 20mg PO OD x11d	<u>Moderate to Severe</u> • Primaquine* + Clindamycin IV • Pentamidine IV <u>Mild to Moderate</u> • Dapsone* + TMP • Primaquine* + Clindamycin PO • Atovaquone 750mg PO BID

Severity:

- Mild – Moderate PJP: PaO2 >% 70 or A-a gradient <35mmHg
- **Moderate – Severe PJP: PaO2 <% 70 or A-a gradient ≥35mmHg**

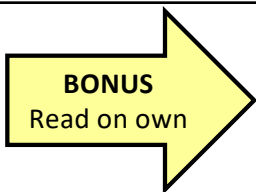
* Check G6PD prior to using dapsone or primaquine
 Note: Septra is considered probably safe for G6PD and CAN be used (though most providers avoid it)

* Add leucovorin when using pyrimethamine

HIV – OI Treatment (2)

OI	Preferred Rx	Alternative Rx
Toxoplasma	Sulfadiazine + pyrimethamine (AI) x 6wk (acute infection) +/- chronic maintenance if ongoing clinical or radiographic disease	• Pyrimethamine + clindamycin (AI) • TMP-SMX 10mg/kg/day (BI) • Atovaquone + sulfadiazine • Atovaquone + pyrimethamine • Atovaquone
MAC	Clarithromycin + ethambutol OR Azithromycin + ethambutol x12 mos	Consider adding rifabutin, amikacin, FQ if advanced HIV or severe disease

- * For toxoplasma – we do not have preferred treatment option in Canada.
- * Check G6PD prior to using dapsone or primaquine
- * Add leucovorin when using pyrimethamine



HIV – Complication Summary (1)

Check out aidsinfo.nih.gov/ for more information on managing other infections in HIV.

CD4 Count	Examples
> 500	• Fever, night sweats, lymphadenopathy, headache, malaise, weight loss • TB (can occur at any CD4 count)
200 to 500	• Non-invasive Candidiasis • Recurrent mucocutaneous HSV-1, HSV-2, VZV • Oral hairy leukoplakia (associated with EBV) • Cutaneous KS (caused by HHV-8) • Sinusitis, bacterial pneumonia (especially invasive pneumococcus) • Cervical dysplasia, cervical carcinoma in situ • Cardiovascular disease, stroke, CKD • Psoriasis, seborrheic dermatitis, bacillary angiomatosis, molluscum contagiosum • Idiopathic thrombocytopenic purpura

HIV – Complication Summary (2)

BONUS
Read on own

CD4 Count	Examples
< 200	• HIV-associated myelopathy (presents with paraplegia) • Visceral KS • Endemic fungi (Coccidiosis, Histoplasmosis, Blastomycosis) • Non-endemic fungi (aspergillosis, cryptococcus) • PJP (subacute presentation over 2 weeks) • NHL > HL, multiple myeloma, leukemia, anal cancer, liver cancer, cervical cancer, vulvar / vaginal cancer
< 100	• Toxoplasmosis • PML (caused by JC virus)
< 50	• CMV (retinitis, colitis) • MAC • CNS lymphoma, HIV associated neurocognitive disorder (HAND)



HIV in Pregnancy

- Vertical transmission in untreated women is 25-35%, <1% with effective treatment
- All women living with HIV contemplating pregnancy need ARV and undetectable VL prior to conception
 - Goal of ARV= undetectable VL throughout pregnancy
- Intra-partum care
 - Always continue ARV
 - If VL > 1000 copies/mL (or unknown) near delivery, give IV zidovudine (AZT) and recommend scheduled C/S
 - If VL suppressed, no increased risk of vaginal delivery
- Post-partum care
 - If maternal VL suppressed within 4 weeks of delivery: Infant given AZT x 4wks
 - If maternal VL not suppressed at birth, infant given presumptive 3-drug ART
 - **Breastfeeding NOT recommended for mothers living with HIV in [US/Canada]**, as safe alternatives are available but woman who choose to breastfeed can be supported



BONUS
Read on own

ARV in Pregnancy – 2025 Update

	Preferred Regimens	Alternative Regimens
NRTI Backbone	• TDF/FTC or TDF/3TC • TAF/FTC or TAF/3TC	• ABC/3TC • ZDV/3TC
3 rd Agent	• Bictegravir (INSTI) • Dolutegravir (INSTI) • Darunavir with ritonavir (PI) (if history of CAB-LA)	• Raltegravir (INSTI) • Atazanavir with ritonavir (PI) • Efavirenz (NNRTI) • Rilpivirine (NNRTI)

- **BIC/TAF/FTC is now the preferred single-tablet regimen during pregnancy (Biktarvy – single pill combo therapy)**
- Dolutegravir, previously thought to increase risk of neural tube defects, has been shown to have no statistically significant risk of NTD, and is now a preferred drug in pregnancy
- Counsel patients on risks of NTD, with folic acid for all patients

This would be way beyond scope of GIM exam!

Key points: keep on ARV treatment, consult with HIV specialist, OB, Peds, pharmacist etc.

Preventing HIV: PrEP and PEP

Pre-Exposure Prophylaxis (PrEP)

- Up to 100% efficacy in preventing HIV transmission with adherence
- Different guidelines exist: Canada (2017) vs CDC (2025).
- Regimens:
 - TDF + FTC (1 tablet daily, or 2-1-1 for MSM)
 - TAF + FTC (1 tablet daily or 2-1-1 for MSM, not indicated for receptive vaginal intercourse)
 - CAB (second IM dose at 1 month, every 2 months afterwards)
- Monitoring:
 - Initial: HIV serology, Hepatitis A, Hepatitis B, Hepatitis C, Gonorrhea/Chlamydia screen, syphilis serology, Cr, lipid panel, urinalysis, pregnancy test
 - Q3m: HIV serology, Gonorrhea/Chlamydia screen, Syphilis serology, Cr, Pregnancy test
 - Q12m: Hep B (if non immune), Hepatitis C, Lipids

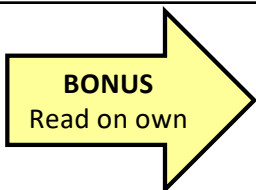
<https://www.cdc.gov/hivnexus/hcp/prep/index.html>

<https://www.cdc.gov/hivnexus/hcp/pep/index.html>

Post-Exposure Prophylaxis (PEP)

- Indication: Moderate-to-high risk exposure to a person with transmissible HIV or of unknown status for moderate to high-risk exposures. Many guidelines exist, eg CDC (2025), Canada (2017)
- Start **as soon as possible**, within 72 hours
- First Line Regimens – all for 28 days:
 - TDF + FTC + DOL
 - TDF + FTC + RAL
- Monitoring:
 - Initial: HIV serology, Hepatitis A serology, Hepatitis B screen, Hepatitis C serology, CBC, ALT, Cr, Pregnancy test
 - Non-occupational: Gonorrhea/Chlamydia screen, syphilis serology,
 - 2 weeks: CBC, ALT, Cr
 - 3 months: HIV serology, Hepatitis C
 - Non-occupational: Gonorrhea/Chlamydia screen, syphilis serology

Doxy PEP: 200 mg x1 dose within 72 hours of sex for gonorrhea, chlamydia, syphilis



Injectable ARV

- Cabotegravir/Rilpivirine only long-acting injectable ARV (q4 or 8 week dosing)
- Treatment: NOT for initiating ARVs in new diagnosis. To qualify for switch to Cabotegravir/Rilpivirine:
 - Virologically suppressed on ARVs for 3-6 months with no treatment failure
 - No resistance to Cabotegravir or rilpivirine
 - Able and willing to attend injections
 - Hep B negative
- Note: Higher rates of failure with HIV-1 subtype A6/A1 and BMI >30
- Oral lead in to assess tolerability is optional, no longer mandatory
- Injection site reactions common and decrease over time
- Injectable Cabotegravir approved for PrEP in Canada in May 2024



HIV and Cardiometabolic risk

- HIV is a chronic condition → both the infection and its treatment can increase risk of cardiovascular illness
 - Persons living with HIV have a 1.5-2X increased risk of CV events compared to HIV-negative controls
 - Medications:
 - Abacavir - ?increased CV events, avoid in patient with high CV risk
 - Protease inhibitors – dyslipidemia, insulin resistance
 - Treat traditional CV risk factors aggressively
- Lipid guidelines (CDC 2025)
 - Framingham / 10yr ASCVD risk >20% = high intensity statin
 - Framingham / 10 yr ASCVD risk 5-20% = moderate intensity statin (rosuva 10, atorva 20)
 - Persons 40-75 yrs with risk <5% = consider moderate intensity statin especially if other CV risk factors or HIV risk (CD4 high, fatty liver, metabolic syndrome, lipodystrophy, Hep C co-infection)

MCQ 2 2026

42 year old man presents to hospital with new diagnosis of HIV and is started on ART. Which of the following would not be a part of the initial assessments in the months following the start of therapy:

- a) Starting a statin if his 10-year ASCVD risk estimates are <5%
- b) Screening for HPV-related anal cancer
- c) Vaccination for HepB if there is no evidence of prior infection or vaccination on serology
- d) **Vaccination with PCV-13 followed by PCV-23 at least 8 weeks later**

The PCV-13 and PCV-23 vaccines have been phased out of PHAC and NACI guidelines in favor of a single dose of Pneu-C-20 or Pneu-C-21 in the last year. (a) is true following the incorporation of the REPRIEVE trial into guidelines as there continues to be emphasis in HIV care on reducing long-term risk of comorbidities, (b) HPV screening is also incorporated now into management, and (c) has long been true

12. Tuberculosis and non-tuberculous mycobacteria

References:

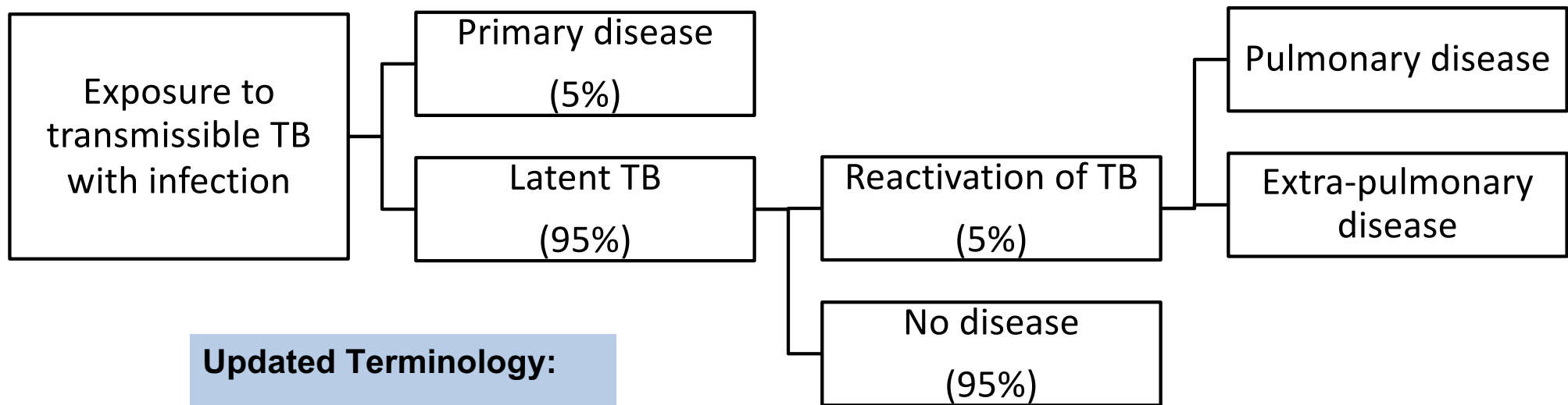
- Canadian TB Standards – 8th Edition (2022).
 - Covers diagnosis and treatment for both latent and active TB
- www.tstin3d.com
 - Great tool to assist in determining likelihood of latent TB based upon TST/IGRA results, epidemiology and BCG status
-  Updates on the Treatment of Drug-Susceptible and Drug-Resistant Tuberculosis: An Official ATS/CDC/ERS/IDSA Clinical Practice Guideline
<https://doi.org/10.1164/rccm.202410-2096ST>
- <http://bcgatlas.org> – Determining likelihood of BCG vaccination based on geographic origin

MCQ 3 2026

A 37 year old woman who has newly immigrated to Canada from a TB-endemic country is found to have a positive TST, although an IGRA performed afterwards is negative. Which is not a possible reason for this discordant result?

- a) The patient had been given repeated doses of the BCG vaccination after 12 months of age.
- b) The wrong TST threshold was used to determine the result
- c) The patient had a single dose of the BCG vaccine at birth
- d) The IGRA sample was incorrectly stored prior to processing

TB Exposure Risk



Updated Terminology:

Latent TB = TB Infection

Active TB = TB Disease

Latent TB – Diagnosis

- Two accepted tests: TST and IGRA
- Neither can separate LTBI from active TB

Decision to test= decision to treat – only check if high risk/occupationally indicated

TST – Tuberculin Skin Test

- In vivo
- May be affected by BCG after 1 year old
- 2 patient visits
- Inter-reader variability
- Sensitivity 80-90%; Specificity >95% if no BCG, 60% with BCG
- Cheap - use if repeat testing required
- **May be (-) if immunosuppressed**

IGRA – Interferon Gamma Release Assay

- In vitro
- Not affected by BCG - use if prior BCG
- 1 patient visit - use if unlikely to return
- Minimal inter-reader variability
- Sensitivity 80-90%; Specificity >95% with or without BCG history
- Often not covered (\$\$ to patient)
- **May be (-) if immunosuppressed**



Latent TB – Diagnosis

TST Result	Situation in which reaction is considered positive
0-4 mm	Generally considered negative
5 mm (higher pre-test probability)	People living with HIV
	Contact with infectious TB case within past two years
	Presence of fibronodular disease on CXR (evidence of healed, untreated Tb)
	Prior to Organ transplantation (ie. Immunosuppressive rx)
	Planned Biologic use including TNF-alpha inhibitor use <u>or DMARDs</u>
	Planned other immunosuppressive drugs (incl corticosteroids Prednisone ≥ 15 mg per day for at least 1 mon.)
≥ 10 mm	Stage 4 or 5 CKD (with or without dialysis)
	All other situations (ie. <u>DM even if well controlled</u>, malnutrition (<90% ideal body wt), silicosis, hematologic malignancies, certain carcinomas (Head and Neck, Lung and/or GI tract), <u>cigarette smoking, alcohol consumption > 3 drinks/d</u>), TST conversion (within 2 years)

BOLD = Very High or High reactivation risk

Underline = moderate reactivation risk

TB Infection – Treatment (Canada 2022)

1. Interpret TST (Positive Predictive Value)
 2. Exclude active disease
 3. Determine risk of re-activation
 4. Consider risk of AEs
- } Age-Dependent



DRUG	SCHEDULE	DURATION	CONSIDERATIONS
<u>FIRST LINE REGIMENS</u>			
Rifampin (4R)	Daily	4 Months	Rash, Drug Interactions
Rifapentine + Isoniazid (3HP)	Weekly	3 Months	Flu-like effects, drug interactions
<u>SECOND LINE REGIMEN</u>			
<i>Isoniazid (9H)</i>	<i>Daily</i>	<i>9 Months</i>	<i>Hepatotoxicity, peripheral neuropathy</i>
<u>ALTERNATIVE REGIMENS</u>			
Isoniazid (6H)	Daily	6 Months	Hepatotoxicity, peripheral neuropathy
Isoniazid	Twice Weekly	9 Months	Hepatotoxicity, peripheral neuropathy
Isoniazid + Rifampin (3HR)	Daily	3 Months	Hepatotoxicity, peripheral neuropathy, Drug interactions

TB Disease Treatment (Canada 2022)

Standard Regimen for Suspected* Drug Susceptible TB

	INTENSIVE PHASE (2 Months)	CONTINUATION PHASE (min 4 Months)
INH	5 mg/kg (up to 300mg)	
RMP	10 mg/kg (up to 600mg)	
PZA + vitB6	20-25 mg/kg	
EMB	15-20 mg/kg	

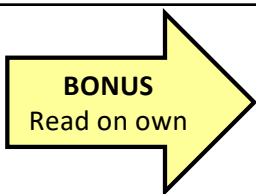
- **Intensive Phase** = 3 or 4 active drugs for rapid killing (usually start RIPE, then drop EMB if susceptible)
- **Continuation Phase** = at least 2 active drugs, duration dependent on regimen, adherence, site of infection, response
- Add B6 (pyridoxine) to prevent peripheral neuropathy
- Add Steroids for TB meningitis or pericardial disease

ATS/CDC/ERS/IDSA guideline update (Dec 2024)

New first line option for drug susceptible disease:
2 months INH, rifapentine, PZA, moxifloxacin, then
2 months INH, rifapentine, and moxifloxacin



For rifampin-resistant disease, 6 months of:
BPaL – Bedaquiline, Pretomanid, Linezolid OR
BPaLM - Bedaquiline, Pretomanid, Linezolid, Moxi



Active TB - Side Effects

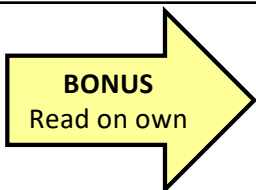
Isoniazid (INH) - rash³, hepatitis², neuropathy
Rifampin (RMP) - drug interaction, rash¹, hepatitis³
Pyrazinamide (PZA) - hepatitis¹, rash², arthralgia
Ethambutol (EMB) - eye toxicity, rash⁴

* Number implies likelihood
out of all first-line TB meds

Neuropathy Prevention:
50 mg VitB6 in at risk patients
(etoh, DM, seizure, malnutrition,
HIV, CKD, all pregnant and
breastfeeding women... practically
most programs put everyone on b6)

If neuropathy develops: 100 mg
daily

	Common adverse events	Uncommon but important adverse events	Rank for probability of hepatitis*	Rank for probability of rash
First-line drugs				
INH	Rash, hepatitis, neuropathy	CNS toxicity, anemia	2	3
RMP	Drug interactions, rash	Hepatitis, 'flu-like illness, neutropenia, thrombocytopenia	3	1
PZA	Hepatitis, rash, arthralgia	Gout	1	2
EMB	Eye toxicity	Rash	4	4



Active TB in Pregnancy

- Elevated risk of TB disease and significant associated morbidity to both woman and fetus
- Risk of untreated active TB to a pregnant woman and her fetus outweighs risk of adverse effects of drugs used in its treatment
- INH, RIF, EMB are considered **SAFE** in pregnancy, so all three should be used as initial treatment
 - Category C due to lack of controlled studies in pregnancy but lack track records of safety in pregnancy
- PZA added if extensive disease, smear-positive pulmonary disease, disseminated TB, intolerance of other agents
 - No formal studies on safety in pregnancy but no reports of teratogenicity
 - WHO recommends use in pregnancy
- Do not treat **latent** TB until after delivery unless high risk of TB reactivation, and use 4R regimen

Latent TB and HIV

- Anyone with TB should have an HIV test (and vice versa)
 - HIV confers a significant increase in risk of reactivation, up to 5% per year (compared to 5% lifetime risk for those living without HIV)



- Preferred regimens:
 - 3HP: Weekly INH + Rifapentine for 3 months
 - 3HR: Daily INH + Rifampin for 3 months
 - (Alternative: INH x 6-9 months OR Rifampin for 4 months)
- * May need to adjust ART for drug interactions with Rifamycins

Active TB in persons living with HIV

- With lower CD4 counts, CXR findings are different (e.g. increased cavitation) and extrapulmonary disease is more common
- Treatment is the same as for TB patients living without HIV
- ART must be adjusted for interactions with Rifamycins
 - Initiation of ART in ART-Naïve Patients:
 - If CD4 < 50 → within 2 weeks
 - If CD4 > 50 → within 8 weeks
 - Pregnancy → ASAP regardless of CD4
 - If TB meningitis → defer for ~8 weeks given ↑ risk of IRIS (especially if low CD4 count)
- IRIS: Increased risk if low CD4, high viral load, disseminated TB
 - Pre-emptive steroids if high risk for IRIS
- Steroids for TB meningitis (similar to patients without HIV) but not TB pericarditis

Guidelines for the Prevention and Treatment of Opportunistic Infections in Adults and Adolescents with HIV: Mycobacterium tuberculosis Infection and Disease. <https://clinicalinfo.hiv.gov/en/guidelines/hiv-clinical-guidelines-adult-and-adolescent-opportunistic-infections/mycobacterium>

Non-tuberculous Mycobacteria

NTM infection is one cause of a false positive Tuberculin Skin Test, but not an IGRA!

Over 190 species and subspecies: *M. avium* complex, *M. kansasii*, *M. xenopi*, *M. abscessus*

Diagnostic criteria (NTM pulmonary disease):

BOTH Clinical (pulmonary or systemic sx) **AND** Radiologic (nodular or cavitary lesions on CXR, or CT with bronchiectasis) **PLUS** one of:

- 2 or more sputum positive for same species NTM
- 1 BAL culture positive for NTM
- Biopsy with mycobacterial histology (AFB/granuloma) and positive culture

Treatment:

- Requires discussion around clinical factors, infecting species and patient priorities.
- Ideally susceptibility-based therapy (not empiric)
- Minimum 3-drug regimen (macrolide, ethambutol, +/-rifampicin +/- aminoglycoside)

MCQ 3 2026

A 67 year old woman who has newly immigrated to Canada from a TB-endemic country is found to have a positive TST, although an IGRA performed afterwards is negative. Which is not a reason for this discordant result?

- a) The patient had been given repeated doses of the BCG vaccination after 12 months of age.
- b) The wrong TST threshold was used to determine the result
- c) The patient had a single dose of the BCG vaccine at birth**
- d) The IGRA sample was incorrectly stored prior to processing
- e) Pulmonary infection with *Mycobacterium avium*

People who have been vaccinated with BCG before 1 year of age generally do not have false-positive results to TST testing after 10 years old (<1%). For this patient that is now 67, the chances of this being a false-positive from the vaccine is exceedingly unlikely.

13. Travel & Tropical Infections



References

- Canadian Recommendations for the Prevention and Treatment of Malaria. An Advisory Committee Statement (ACS) committee to advise on tropical medicine and travel (CATMAT). PHAC 2024.
- Taylor SM, Molyneux ME, Simel DL, *et al*. Does this patient have malaria? *JAMA* 2010;304:2048-56.
- Zika Working Group on behalf of the Committee to Advice on Tropical Medicine and Travel (CATMAT). Canadian recommendations on the prevention and treatment of Zika virus: Update. March 2020.

Fever in Returned Traveler

BONUS
Read on own

Pre-Travel

- Past medical history, medications
- History of a pre-travel consultation
- Travel immunizations (Hep A/B, yellow fever, typhoid, rabies, meningitis, influenza)
- Adherence to malaria chemoprophylaxis

Travel

- Purpose of travel (Visiting friends and relatives (VFR) greatest risk)
- Itinerary (dates/locations, incl. layovers), season
- Type of accommodations (urban versus rural)
- Insect precautions taken (repellant, bed nets)
- Individual exposures
 - raw meat, seafood, street foods, unclean water, unpasteurized dairy products
 - Freshwater exposure (e.g. swimming, rafting)
 - Visits to farms, slaughterhouses, funerals, hospitals
 - Animal bites, insect or arthropod bites, hiking/caving
 - Body fluid exposures (e.g. tattoos, **sexual activity**)
- Medical care while overseas

Post-Travel

- Timing of fever / clinical symptoms in relation to international travel
 - Short incubation periods < 2 weeks = **Malaria**, Dengue, Chikungunya, traveler's diarrhea, viral URTI, influenza
 - Longer incubation periods > 2 weeks = **Malaria**, TB, hepatitis, HIV, enteric fever due to *Salmonella* spp.
- Pattern of fever
 - Daily: **Malaria**, traveler's diarrhea, viral RTI, enteric fever
 - Biphasic: **Malaria**, Dengue
 - Relapsing: **Malaria**, enteric fever

Fever in Returned Traveler

Differential Diagnosis

- Infectious – Travel-related
 - **Malaria, malaria, malaria!!!**
 - Typhoid /enteric fever
 - Dengue, Chikungunya, Zika virus
 - Viral hepatitis
 - Others: Rickettsiae, Brucellosis, Leptospirosis, Q-fever, Ebola
 - STIs, acute HIV
- Infectious – Non-travel related
 - UTI, CAP, mono, meningitis, *C. diff*, influenza, COVID-19
- Non-infectious (travel or non-travel)
 - VTE, drug fever, illicit drugs S/E

Investigations

- CBC + diff, peripheral smear
- Electrolytes, Cr, urea, glucose
- ALT, AST, ALP, Bili, INR/PTT
- Blood cultures x2, urine culture
- CXR
- Malaria smears x3 (over 24h)
 - Rapid Diagnostic Tests (RDT e.g. Binax)
- +/- Stool C&S; Stool O&P; *C. difficile*
- +/- Dengue/Chikungunya/Zika PCR or serology
- +/- Hepatitis serology
- +/- HIV, STI screen
- +/- NP swab for viral PCR (including SARS-COV2 and Influenza)

Malaria – Diagnosis

CATMAT, 2024

Species:

- *Plasmodium falciparum*
 - Can be severe
 - Present within 3 months
 - Infects RBCs of all stages
- *P. ovale* and *P. vivax*
 - May present years later due to hypnozoites in liver
- Others:
 - *P. malariae*
 - *P. knowlesi*
 - *P. simium*

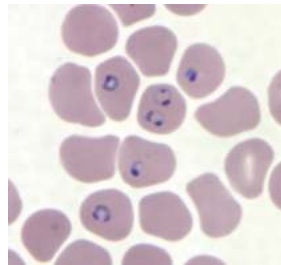
Diagnostic Techniques:

Thick and thin blood smear x3, separated by at least 6 hours over 24 hour period

- Thick smear: Looks for any parasite (sensitivity 87%)
- Thin smear: Identifies parasitemia (%) and speciation by stain

Rapid detection test (RDT)

Separate tests for *P. falciparum* and others; highest Sn for *P. falciparum*



Malaria – Severity Criteria

Criteria for Severe Malaria:

Essentially any end organ dysfunction

- Neurologic = confusion, prostration (severe weakness), seizures
- Respiratory = ARDS, pulmonary edema
- Hematologic = DIC (anemia, thrombocytopenia, elevated LDH), jaundice, hemoglobinuria → Black water fever

If Severe:

1. **Admit to ICU**
2. Start treatment (see next slide)
3. Supportive Care
4. Serial Glucose Checks

CATMAT, 2020

- Severe anemia (Hgb < 50)
- Hypoglycemia (glucose < 2.2)
- Acidosis (pH < 7.25, HCO₃ < 15)
- Renal impairment (Cr > 265)
- Lactic Acidosis
- Hyperparasitemia
 - ≥5% for non-immune adults
 - Non immune = born in non-endemic countries or who have moved out of endemic country >6-12 mos
 - ≥10% for semi-immune adults
 - Semi-immune = birth and long term residence in an endemic country and prior episodes of malaria.

Malaria – Management

CATMAT, 2024

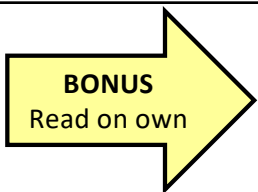
Severity	Species	Resistance	Treatment
Uncomplicated	<i>P. falciparum</i>	CS or CR	Atovaquone-proguanil <i>OR</i> Quinine + doxycycline <i>OR</i> Quinine + clindamycin
(can treat with oral meds)	Non-falciparum spp.	CS (most common)	Chloroquine*
		CR (Papua New Guinea)	Atovaquone-proguanil <i>OR</i> Quinine + doxycycline
Complicated	N/A	N/A	IV artesunate X 48h then PO • Atovaquone-proguanil <i>OR</i> • Doxycycline + quinine <i>OR</i> • Clindamycin + quinine (alternate)

Unsure if CS on MCQ?

Based on local epi but generally for *P. falciparum* assume resistant and for non-falciparum assume sensitive (exception is New Guinea)

- * If *P. vivax* or *P. ovale*, add primaquine (check G6PD level first) to treat hypnozoite stage
- Repeat smears q6-12h to monitor parasitemia
- If artesunate not available for complicated malaria, give IV quinine as alternative (monitor QT and for hypoglycemia)

CS = chloroquine sensitive; CR = chloroquine resistant



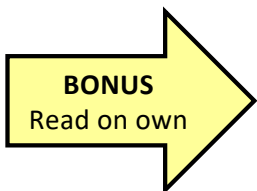
Malaria – Chemoprophylaxis

CATMAT, 2024

Drugs	Notes
Atovaquone-proguanil (Malarone)	Daily dosing; start 1-2d prior and stop 7d post. Insufficient data regarding use in pregnancy. Well-tolerated
Doxycycline	Daily dosing; start 1-2d prior and stop 4w post. GI upset, photosensitivity. Contraindicated in pregnancy
Mefloquine	Weekly dosing; start 1+w prior and stop 4w post. Associated with bizarre dreams and can cause severe neuro-psychiatric symptoms (sz/psychosis); QT prolongation. Safe in pregnancy ; check to ensure no resistance in destination
Primaquine	Alternate option - Use only if others contra-indicated. Daily dosing; start 1-2d prior and stop 7d post Need to check G6PD level. Contraindicated in pregnancy/breastfeeding
Chloroquine/ Hydroxychloroquine	Alternate option - Only use in sensitive areas (e.g. Haiti, DR, central America, etc.). Weekly dosing; start 1w prior and stop 4w post. Safe in pregnancy

Some other infections to know...

Dengue (mosquito-borne)	Zika virus (mosquito-borne)	Chikungunya (mosquito-borne)	Typhoid (Salmonella typhi/ paratyphi)	Leptospirosis (animal waste → soil, water exposure)
Incubation: < 2 wks Features: fever, maculopapular rash, retro-orbital pain, myalgias (“break bone” fever), cytopenias (<u>thrombocytopenia</u>). Rx: supportive care; avoid NSAIDs	Incubation: < 2 wks Features: fever, rash, arthralgias, myalgias, conjunctivitis, HA, retro-orbital pain; association with microcephaly and GBS Rx: supportive care	Incubation: < 2 wks Features: fever, polyarthralgia (NOT arthritis usually), lymphopenia, maculopapular rash Rx: supportive care	Incubation: 5-21 d Features: fever, flu-like illness, salmon-coloured spots, constipation, abdominal pain, relative bradycardia Rx: IV ceftriaxone, ciprofloxacin or azithromycin *increasing resistance with FQs in SE Asia	Incubation: 2-26 d, ~10 d Features: fever, myalgias, HA, conjunctivitis, hypokalemia, cytopenias, sterile pyuria. Rare severe: jaundice + renal failure (Weil’s dz.), ARDS, pulm. Hemorrhage. Rx: Mild disease: doxycycline or azithromycin Severe: IV ceftriaxone, Penicillin or doxycycline Supportive care



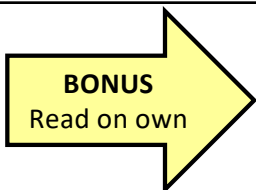
Viral Hemorrhagic Fevers

- Group of severe diseases caused by 4 virus families:
 - Arenaviridae (E.g. Lassa Fever)
 - Filoviridae (E.g. Ebola, Marburg)
 - Bunyaviridae (E.g. Crimean-Congo hemorrhagic fever)
 - Flaviviruses (E.g. Yellow Fever, Dengue) – arthropod transmission
- All VHFs can progress to severe disease but likelihood varies
- Severe disease usually hemorrhagic or vascular leak manifestations
- Natural hosts are animals or arthropods and outbreaks occur where host is present
- Several outbreaks over the world in recent years
- **Important thing is to alert IPAC immediately if symptomatic patient arrives who travelled to outbreak zone in past 21 days and start droplet/contact + N95 precautions (except flaviviruses)**

Current worldwide Outbreaks:

Sept 2024 - ? = Rwanda, Marburg

Sept 2022 – Jan 2023 = Sudan, Ebola (declared over Jan 2023)

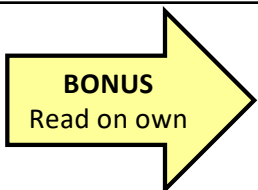


Ebola Virus

[CATMAT, 2023](#)
[PHAC, 2023](#)

- Incubation: 2-21 days (avg 8-12)
 - Most recent Uganda Outbreak Sept 2022-Jan 2023
 - Prior outbreaks Dem. Rep. of Congo, Guinea, West Africa
- Syndrome:
 - Fever, myalgias → GI symptoms, anorexia
 - Bleeding < 20%
- Infection control:
 - call IPAC, droplet/contact with N95
- until further notice
 - Impermeable neck to toe, with N95 and face shield, trained observer
- Diagnosis:
 - Viral culture, NAAT, viral antigens, serology from appropriate sites
- Supportive care
 - Essential procedures/bloodwork only
 - Do not forget DDx -> **MALARIA!**
 - Monoclonal antibody treatments
- Case fatality rate 60%

NB: Investigational vaccine not available in Canada



Measles Virus

- Outbreaks, waning vaccination, airborne transmission
- Incubation:
 - ~14 days post exposure
- Syndrome:
 - fever, cough, coryza, conjunctivitis, Koplik spots, rash (centrifugal)
- Infection Control:
 - Airborne precautions, contact tracing (4 days prior to rash, 4 days after rash)
- Diagnosis:
 - PCR of pharynx/NP/urine; serology can be false negative early
- Complications:
 - Pneumonia, encephalitis, subacute sclerosing panencephalitis (SSPE)
- Post-exposure prophylaxis:
 - MMR, Measles immunoglobulin if susceptible, based on time from exposure

BONUS
Read on own

Parasites Simplified

Protozoa

Amoebae

Acanthamoeba

- Granulomatous Amebic Encephalitis
- Contact lens related Keratitis
- Treatment Uncertain

Naegleria fowleri

- Primary Amebic Encephalitis
- Warm Fresh Water inhalation
- **No Canadian Cases**
- Fatal, No treatment

Entamoeba histolytica

- Luminal disease, liver abscess
- Microscopy (can't differentiate from non-pathogenic), Antigen, PCR
- Paromomycin (luminal) +/- Metronidazole (for systemic)

Apicomplexans

Plasmodium

See Malaria Slides

Babesia

See Bonus Slides

Toxoplasma

See HIV slides

Cryptosporidium

- Fecal-oral
- Self-limited diarrhea in immunocompetent
- Chronic, life threatening in immunocompromise
- Microscopy
- Nitazoxanide
- Reduce Immunosuppression

Flagellates

Trypanosoma

- T. brucei – African Sleeping Sickness
- T. cruzi – Chagas (Central/S. America)

Leishmania

- Sand Fly Vector
- Cutaneous ulcers, mucocutaneous and visceral forms
- Can give topical treatment for mild cutaneous, Amphotericin for others

Giardia

- Fecal-Oral
- Day-care, drinking/swimming in lake or river, MSM
- Diarrheal infection
- Microscopy
- Metronidazole x 14d

Parasites Simplified

Helminths

Flat Worms (Platyhelminths)

Flukes (Trematodes)

Schistosoma
Fasciola
Paragonimus
Opisthorchis
Clonorchis

- Microscopy: O&P x 3 from relevant site
- +/- Serology
- Praziquantel EXCEPT Fasciola (Triclabendazole)

Tapeworm (Cestodes)

Taenia solium
Taenia saginata
Echinococcus
Diphyllobothrium

- Microscopy: O&P x 3 (stool)
- Serology
- Imaging for Echinococcus or Taenia solium
- In gut: Praziquantel
- In tissue: Albendazole
- Echinococcus treatment is dependent on stage (from imaging) but can involve albendazole or surgery/ aspiration

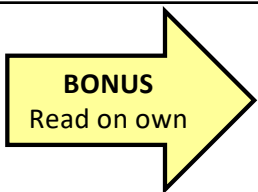
Roundworm (Nematodes)

Ascaris
Strongyloides
Trichinella
Trichuria
Enterobius
Cutaneous LM

- Microscopy: O/P x 3
- Serology
- Some (Cutaneous larva migrans) clinical diagnosis
- In gut: Mebendazole
- In tissue: Albendazole (ivermectin is alternative sometimes)

Microfilaria
W. bancrofti
Onchocerca

Blood Smear



Helminths (worms) 101

- About 300 species that cause human disease, many low yield for exam. Don't waste time studying them all
- Clue is **eosinophilia** (Helminths contained to GI tract may not have eosinophilia).
- Dx with stool/urine/sputum microscopy (ova and parasites), +/- serology
- *Echinococcus (granulosus/multilocularis)*: Liver cysts, sometimes lung cysts. Diagnosed with combo imaging + serology/Antigen. Staged based on imaging, tx depends on stage but may include albendazole, aspiration, surgery or monitoring.
- *Schistosoma (mansoni/haematobium* etc): Water/snail host, subtropical and tropical areas, chronic infection, can cause liver/bladder CA. Dx. Tx. Praziquantel
- *Taenia solium* (pork): Can cause taeniasis (eating infected/undercooked meat), or neurocysticercosis (eating eggs). Tx neurocysticercosis with albendazole +/- praziquantel +/- steroids
- *Trichinella spiralis* (trichinosis) from eating undercooked wild animal meat (bear/pork). GI symptoms, muscle pain (cysts). Tx. Albendazole/mebendazole.

<https://www.cdc.gov/parasites/az/index.html>

Strongyloides stercoralis

- From skin contact with soil in tropical and subtropical regions (e.g. walking on a beach) – assume exposure in anyone living for >1 month in endemic areas
- Can cause lifelong asymptomatic infection where host continuously reinfects self
- Symptomatic infection may be eosinophilia, diarrhea, itching and respiratory symptoms (may diagnose with O/P or serology)
- With glucocorticoids or other immunosuppression can get disseminated disease which can cause recurrent gram-negative infections (including meningitis) and is highly fatal
- Screen exposed patients with serology prior to starting immunosuppression
- Treatment is Ivermectin (1 or 2 doses)

14. Fungal Infections

References:

- Pappas PG, Kauffman CA, Andes D, *et al*. Clinical practice guidelines for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2016.
- Practice Guidelines for the Diagnosis and Management of Aspergillosis: 2016 Update by the Infectious Diseases Society of America. *Clin Infect Dis* 2016 Aug 15;63(4):e1-e60. doi: 10.1093/cid/ciw326. Epub 2016 Jun 29.
- Chapman SW, Dismukes WE, Proia LA, *et al*. Clinical practice guidelines for the management of blastomycosis: 2008 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2008;46:1801-12.
- Wheat LJ, Freifeld AG, Kleiman MB, *et al*. Clinical practice guidelines for the management of patients with histoplasmosis: 2007 update by the Infectious Diseases Society of America. *Clin Infect Dis* 2007;45:807-825.
- Galgiani JN, Ampel NM, Blair JE, *et al*. 2016 Infectious Diseases Society of America (IDSA) Clinical Practice Guideline for the Treatment of Coccidioidomycosis. *Clin Infect Dis*. 2016;63(6):e112-e146. doi:10.1093/cid/ciw360

Candidemia

Species	Fluconazole (IV=PO)	Echinocandins (not for CNS/eyes)	Amphotericin B (liposomal preferred)
<i>C. albicans/dubliniensis/tropicalis</i>	S	S	S
<i>C. glabrata</i>	Variable**	S	S
<i>C. krusei</i>	R	S	S
<i>C. parapsilosis</i>	S	Variable***	S
<i>C. lusitaniae</i>	S	S	R

**Higher rates of resistance to Fluc. When susceptible, requires higher dose

***Tends to have higher MICs to echinocandins

Risk Factors:

- Use of broad-spec antibiotics
- ICU admission
- CVC
- TPN
- Neutropenia
- Immunosuppressive agents (steroids, chemo, etc.)
- Intra-abdominal surgical procedures
- Necrotizing pancreatitis
- Candida colonization x 3 sites

Management:

- Stable, no recent azole exposure → fluconazole
- Unstable, neutropenic, or recent azole exposure → echinocandin
- **Pregnancy → amphotericin B**
- **CNS infections → amphotericin B +/- flucytosine**

Remove central lines (especially if prolonged candidemia)
Treat for two weeks from first negative blood culture (if no metastatic focus)

* Consult ophthalmology to r/o endophthalmitis

* Consider TTE or Abdo US if persistent candidemia to rule out IE

Pappas PG et al, 2016

Aspergillus

Syndrome	Additional Details and Treatment
ABPA/Allergic Rhinosinusitis	Hyper-sensitivity response to Aspergillus allergens/precipitans Elevated IgE, asthma, brown sputum, eosinophilia, bronchiectasis Rx: Steroids/anti-IgE +/- itraconazole (anti-fungal controversial)
Aspergilloma	Mycetoma that forms within a pre-existing cavity Solitary lesion = Rx surgical resection +/- antifungal Multiple lesions = Rx antifungal x 6 months
Chronic Cavitary Pulmonary Aspergilosis (CCPA)	Pre-existing structural lung disease (most commonly COPD) → progression of above Weight loss, worsening cough +/- hemoptysis Rx antifungal x 6 months
Invasive aspergillosis	Opportunistic infection seen in neutropenia/cellular immunocompromise Diagnosis using imaging (CT chest), indirect tests (galactomannan of serum and sputum), or direct tests (fungal culture or pathology) Rx: Voriconazole x ≥6 weeks or longer

CID, 2016

Dimorphic/(Endemic) Fungi

BONUS
Read on own

Dimorphic Fungi	Notes and Treatment
<i>Blastomyces dermatitidis</i> ie. Blastomycosis Chapman SW et al, 2008	• Border of Great Lakes (incl. Northern Ontario), Southeastern/central USA, and St. Lawrence River (now sometimes seen in GTA) • Pneumonia, skin/joint infections • Dx: Fungal cx, PCR, path, urine/serum antigen • Rx: Itraconazole (mild to moderate), Amphotericin B (severe) • Duration 6-12 months, or 3 months after complete resolution
<i>Histoplasma capsulatum</i> ie. Histoplasmosis Wheat LJ et al, 2007	• St. Lawrence, Ohio rivers • Self-limited pneumonia, M.tb. mimicker • Dx: Fungal cx, PCR, path, urine antigen • Rx: None (mild), Itraconazole (moderate), Amphotericin B (severe) • Duration ~12 weeks
<i>Coccidioides spp.</i> ie. Coccidioidomycosis Galgiani JN et al, 2016	• New Mexico, Arizona, Mexico, Central America (valley fever) • Suspect in returned traveler with pneumonia, meningitis • Dx: Fungal cx, path, antigen; LP if symptoms • Rx: None (asymptomatic), Itraconazole (symptomatic)

Questions?

DON'T FORGET CHECK OUT THE BONUS MATERIAL FOR MORE READING ON

- Extra material on:
 - AMMI Canada Duration of Antibiotics Practice Point
 - Emerging pathogens
 - Lyme Disease and Tick Borne infections
 - Fever of unknown origin
 - Infections in immunocompromised hosts
 - Nosocomial infections
 - IPAC
 - Exposure management (bites, needlesticks)
 - Adult immunizations including vaccine primer
 - Choosing Wisely

****Bonus questions available online!**

BONUS
Read on own

AMMI Duration of Abx: GU Infections

Syndrome	Duration of Treatment	Comments
Acute Simple Cystitis	Nitrofurantoin 100mg BID x 5 days	Young non-pregnant women or young adults with normal GU anatomy and normal renal function
	TMPSMX 1 DS BID x 3 days	
	Fosfomycin 3g PO X 1	
Complicated UTI OR Pyelonephritis (Oral)	Quinolone or Beta Lactam x 7 days	Excludes patients with urogenital abnormalities or stents. Duration depends on rapidity of improvement

SOURCE: ASSOCIATION OF MEDICAL MICROBIOLOGISTS AND INFECTIOUS DISEASE CANADA Grant, J et al. JAMMI 6(3); 2021: 181-197
Practice Point “Shorter is better” – endorsed by med micro and ID docs of Canada

BONUS
Read on own

AMMI Guidelines: Respiratory Tract

Syndrome	Duration of Treatment	Comments
Strep pharyngitis	Pen V x 10 days	Most data from peds, they still say 10 days here on guideline.
Acute Otitis Media or sinusitis , uncomplicated	5 days	Should meet diagnostic criteria Doesn't include Complicated sinusitis eg epidural or orbital abscess
Community Acquired Pneumonia	5 days	Excludes immunocompromised or underlying lung disease ex emphysema. Pt must be improving w normal vital signs for 2 d
HAP/VAP	7 days	Generally. RCT of 8 vs 15 d showed no difference in important outcomes, pts receiving less antibiotics had less re-infection w resistant organisms. IDSA recommends 7 d for VAP.
AECOPD	5-7 days	5 days for most if they meet criteria for antibiotics

BONUS
Read on own

AMMI Guidelines: Skin Bone GUT

Syndrome	Duration of Treatment	Comments
Uncomplicated cellulitis	5 – 7 days	Usually due to GAS or S aureus
Acute vertebral osteomyelitis in adults	6 weeks	Not associated w implantable device; Assumes S aureus is pathogen but Rx duration could be longer if Salmonella or Brucella infection.
Adults with native joint osteoarticular infections (septic arthritis)	2 weeks if small joint after surgical drainage 4 weeks if small joint after surg drainage	Only applies to postop pts w causal organism and susceptibility profile
Bowel Perforation: Traumatic→ Gastro-duodenal (eg PUD)→	No more than 24h postop	Operated on within 12h of trauma Operated on within 24h of perforation
Intraabdominal infection	<7 days	With source control with adequate drainage in place.

BONUS
Read on own

AMMI Guidelines: Bacteremia

Syndrome	Duration of Treatment	Comments
Gram Neg Enterobacteriaceae such as <i>E coli</i>	7 days	Assumes source control (e.g. drainage of abscess)
<i>Staph aureus</i> Bacteremia: Uncomplicated	14 days IV	ID consult recommended; ensure no endocarditis, osteomyelitis or other site infection
<i>Staph aureus</i> Bacteremia: Complicated	4-6 weeks IV	ID Consult recommended; endocarditis, metastatic foci of infection, persistent bacteremia, etc.
<i>Strep pneumonia</i> associated w clinical pneumonia	5-7 days	Must show clinical improvement and meet criteria for discontinuing abx for pneumonia

BONUS

Read on own

Tick-borne Disease

References:



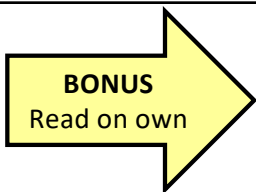
- Lantos PM, Rumbaugh J, Bockenstedt LK, *et al*. Clinical Practice Guidelines by the Infectious Diseases Society of America (IDSA), American Academy of Neurology (AAN), and American College of Rheumatology (ACR): 2020 Guidelines for the Prevention, Diagnosis, and Treatment of Lyme Disease. *Arthritis and Rheumatology* 2020: 1-9. DOI 10.1002/art.41562
- Krause PJ, Auwaerter PG, Bannuru RR, *et al*. Clinical Practice Guidelines by the Infectious Diseases Society of America (IDSA): 2020 Guideline on Diagnosis and Management of Babesiosis [published correction appears in *Clin Infect Dis*. 2021 Jul 1;73(1):172-173]. *Clin Infect Dis*. 2021;72(2):e49-e64. doi:10.1093/cid/ciaa1216
- Lyme Disease. Public Health Agency of Canada. 2023. <https://www.canada.ca/en/public-health/services/diseases/lyme-disease/health-professionals-lyme-disease.html>
- Diseases Transmitted By Ticks. CDC. 2023. <https://www.cdc.gov/ticks/diseases/index.html>

BONUS

Read on own

Arthropod Infections in Canada

- **Ixodes scapularis**
 - Lyme (*Borrelia burgdorferi*)
 - Anaplasmosis
 - Babesiosis
 - Powassan Virus
 - *Borrelia miyamotoi*
- **Ixodes pacificus**
 - Lyme (*Borrelia burgdorferi*)
 - *Borrelia miyamotoi*
- **Other Ticks**
 - Tick Borne Relapsing fever
 - Rocky Mountain Spotted Fever
 - Tularemia
 - Colorado Tick Fever
- **Lice**
 - Louse Borne Relapsing Fever
 - *Bartonella quintana*
- **Fleas**
 - *Bartonella henselae*
- **Mosquitos**
 - West Nile Virus
 - Western equine Virus
 - Eastern Equine Virus
 - St. Louis Virus



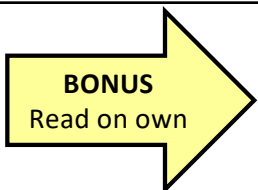
Lyme Disease – Dx and Rx

Lantos et al, 2020

- DO NOT TEST ASYMPTOMATIC PATIENTS
- For patients with early localized infection with erythema migrans, testing is not required (serology is often negative at this early stage) and they can be treated based on the clinical diagnosis
- For patients with signs of early localized infection without erythema migrans, or objective symptoms concerning for disseminated Lyme, testing is recommended using a two-step pathway

Stage	Syndrome	Presentation	Rx
	High risk bite	Confirmed Ixodes spp. In highly endemic area, attached >36 hours	Doxycycline 200mg PO x1 within 72 hrs of tick removal*
Early Localized	Erythema migrans	“target lesion” at site of tick bite	Doxycycline X 10d; Amoxicillin or Cefuroxime x 14d
Early Disseminated	Multiple Erythema Migrans	Multiple target lesions	Doxycycline X 10d; Amoxicillin or Cefuroxime x 14d
	Early neuroborreliosis	CN palsy, Meningitis, radiculopathy, neuropathy	Doxycycline, ceftriaxone, Cefotaxime, pen G X 14-21d
	Carditis	Heart block, myocarditis, pericarditis	Ceftriaxone step down to oral once well Total X 14-21d
Late	Arthritis	Mono/oligo/polyarthritis	Doxycycline X 28d
	Late neuroborreliosis	Encephalopathy, encephalitis, peripheral neuropathy	Ceftriaxone X 14d

*Doxycycline single dose prophylaxis okay for pregnant women, with risk-benefit discussion



Lyme Disease Management Pearls

- **Erythema Migrans:** Typical Lesions are clinical diagnosis. If atypical, acute and convalescent serology
- For early disseminated and late manifestations, serology for diagnosis rather than PCR (including neuro and arthritis)
- Screen with ECG if early Lyme disease and symptoms of carditis (dyspnea, edema, palpitations, lightheadedness, chest pain, and syncope)
- Indications for Admission: significant PR prolongation (PR > 300 milliseconds), other arrhythmias, or clinical manifestations of myopericarditis
 - For telemetry, may require temporary pacing
- Post Treatment Lyme Disease Syndrome: persistent symptoms after treatment. Usually, non-infectious.
 - Clinical signs of ongoing infection (e.g. arthritis): consider second course of tx with IV
 - If ongoing arthritis after 2nd tx course: refer to rheumatology for DMARDs
 - If no arthritis or objective signs of infection: recommend against further antibiotics
- Consider coinfection (Babesia, Anaplasma), especially if ongoing fevers while on antibiotics

BONUS

Read on own

Other Ixodes Infections

Anaplasmosis

- Rickettsial Bacteria: *Anaplasma Phagocytophilum*
- Geography: South Central and eastern Canada
- Incubation: 5-14 days
- Manifestations:
 - fever, malaise, myalgia, and headache
 - DIC, Rhabdo, Resp compromise, hemorrhage
- Diagnosis: Serology +/- PCR
- Treatment: Doxycycline x 7-10 days

Babesiosis

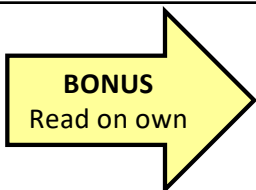
- Protozoa: *Babesia Microti*
- Geography: South central and eastern Canada
- Incubation: 1-4 weeks
- Manifestations:
 - Fever, fatigue, malaise, hemolytic anemia
 - Severe disease associated with severe anemia and multiple complications
- Diagnosis: Blood Smear, PCR, Serology
 - Tetrad (Maltese cross)
- Treatment: Azithromycin + Atovaquone x 7-10 days

BONUS
Read on own

Fever of Unknown Origin (FUO)

References:

- Haidar G, Singh N. Fever of Unknown Origin. *N Engl J Med*. 2022;386(5):463-477. doi:10.1056/NEJMra2111003
- Mourad O, Palda V, and Detsky AS. A comprehensive evidence-based approach to fever of unknown origin. *Arch Intern Med* 2003; 163:545-51.
- Hersch EC and Oh RC. Prolonged febrile illness and fever of unknown origin in adults. *American Family Physician* 2014;90:91-96.



FUO

- Classic Definition: T > 38.3C over 3 wks, with 1 wk investigations
 - Core features are prolonged fever without clear source despite reasonable investigations (rather than rigid definition)
- Differential Diagnosis:

Infection (30%)	Abscess (intra-abdominal) Endocarditis Sinusitis TB and Others – CMV, EBV HIV
Inflammatory (20%)	GCA (age >50) Still's Disease IBD + other seropos
Malignancy (15%)	Lymphoma/leukemia Renal Cell Colorectal
Drugs / miscellaneous (10%)	VTE, Drugs → AEDs, antimicrobials (esp PCN or cephalosporins), NSAIDs, Allopurinol
Idiopathic (25%)	Good prognosis, spontaneous recovery in most pts

BONUS

Read on own

FUO Approach

- **First line investigations:**

- History and physical exam, comprehensive fever diary, travel hx, dietary exposure, animal exposure.
 - Consider hold offending medications (NSAIDs) and Antibiotics (if pt is stable, non neutropenic)
 - Determine need for hospitalization (usually not required!)
- CBC + diff, blood film, lytes, Cr, LDH, TSH, CK, LFTs, ESR/CRP
- Blood cultures (X 2 sets), urine C&S
- HIV serology, +/- EBV, hepatitis, CMV IgM, monospot, Q-fever serology in right setting
- CT Chest/abdo/pelvis or other regions as directed by hx/physical ex. leg dopplers

- **Second line tests:**

- ANA, RF, CCP, ANCA, TSH
- Serologic or PCR testing for tickborne or zoonotic illness as directed by history/exposures
- Tb testing if not done
- Consider biopsy (rashes, nodes, Temporal artery eg (if ESR >50 and patient >50 as directed by history and exam)

*Empiric trials of steroids or antibiotics discouraged as they rarely establish diagnosis

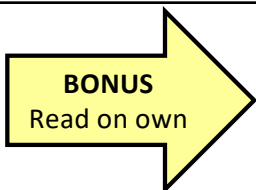
*Reevaluate clinical status, revisit history, consider factitious fever

*FDG PET-CT if available, not done and fevers have not self-resolved

*Consider Metagenomic testing for infection if available (Hairdar G et al, 2022)

*Consider referral for ID/Rheum opinions, query autoinflammatory syndromes (FMF, Schnitzler Syndrome)

Mourad O et al, 2003, Hersch EC et al, 2014, Hairdar G et al, 2022



Infections in Immunocompromised Hosts

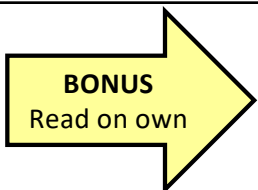
References:

- Fever and Neutropenia in Adults with Cancer: 2018 Update by the Infectious Diseases Society of America; CID 2018 (covered by med onc)
- Shafran et al. Reducing the risk of infection in a 74-year-old man who is to receive prednisone. CMAJ. 2014

BONUS
Read on own

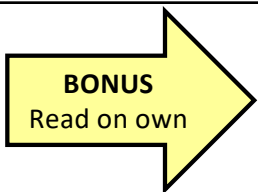
Immunomodulating Therapy Work-up

- **LTBI:** TST/IGRA if >1 TB risk factor* and either TNF-a or Prednisone >15mg/d for >4 weeks (or equivalent).
 - * Close contact w TB, recent immigration high risk country, high risk work/life exposure etc.
 - If positive, rule out active TB and treat latent TB. (can give immunosuppression once treatment started)
- **Hepatitis B:** Screen w/ HBsAg (+/- coreAb) if biologic or Prednisone > 7.5mg/d
 - Consider need for treatment/prophylaxis
 - **Rituximab is high risk for reactivation, must start prophylaxis with entecavir or Tenofovir
- **Hepatitis C:** Some risk of reactivation, screen with serology for TNF-a or long-term steroids
- **PJP:** Consider TMP-SMX prophylaxis if Prednisone > 20mg/d for > 4-8 weeks
- **Strongyloides:** Screen with serology if planned immunosuppression in anyone who lived (usually >1m) in tropic or subtropic area.
 - Treatment with/ Ivermectin 200 mcg/kg PO day 1, 14
- ***HIV:** Although no evidence for worsening infection, individuals are usually screened for HIV alongside Hepatitis B and C when starting immunocompromising treatment



Antimicrobial Prophylaxis in Oncology

- Prophylaxis with **Ciprofloxacin** recommended in those at high risk for FN or prolonged profound neutropenia ($> 7d$ and $ANC < 0.1$)
- **Antifungal** (oral triazole or echinocandin) recommended if prolonged and profound neutropenia such as in AML/MDS, HSCT. Not recommended for solid tumour.
- **PJP prophylaxis** recommended if chemotherapy risk of PJP $> 3.5\%$ e.g. those with ≥ 20 mg pred daily for > 1 mo
- HSV positive undergoing allo-HSCT or leukemia induction should get **acyclovir** for prophylaxis
- **NRTI** (e.g. entecavir or tenofovir) if high risk of Hep B reactivation
- Yearly influenza vaccine recommended
- Those with neutropenia from cancer therapy should avoid prolonged contact with environments that have high concentration with fungal spores



Nosocomial Infection Prevention

VAP

- Avoid intubation if possible
- Minimize sedation
- Maintain physical conditioning
- Keep head of bed between 30 to 45 degrees
- Oral care with toothbrush, **no chlorhexidine**
- Early enteral nutrition
- Change ventilator circuit only if visibly soiled/malfunctioning

CLABSI

- Evidence based list of indications
- Provide education to providers
- Daily chlorhexidine baths
- Hand hygiene prior to catheter insertion
- Subclavian $\geq$ jugular $>$ femoral
- Procedure checklist and all-inclusive kit
- Ultrasound guidance
- Use maximal sterile barrier precautions
- Chlorhexidine skin prep
- Appropriate nurse-to-patient ratio
- Chlorhexidine dressings; change q7d
- Disinfect before accessing catheter
- Remove nonessential catheters
- Surveillance for infection

CAUTI

- Hand hygiene; insert catheter using aseptic technique and sterile equipment
- Use smallest catheter possible
- Maintain sterile, continuously closed drainage system
- Keep collection bag below level of bladder
- Empty collection bag regularly
- Daily discussion of indication for catheter (consider alternatives)
- Avoid unnecessary re-insertion

BONUS
Read on own

Infection Control

References:

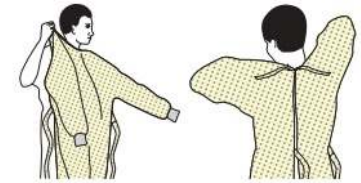
- Routine practices and additional precautions for preventing the transmission of infection in healthcare settings (PHAC, 2013)

Donning/Doffing:

- Droplet Demonstration - 2015 video on website under course materials; available for various clinical scenarios depending on level of PPE required

1. GOWN

- Fully cover torso from neck to knees, arms to end of wrists, and wrap around the back
- Fasten in back of neck and waist



2. MASK OR RESPIRATOR

- Secure ties or elastic bands at middle of head and neck
- Fit flexible band to nose bridge
- Fit snug to face and below chin
- Fit-check respirator



3. GOGGLES OR FACE SHIELD

- Place over face and eyes and adjust to fit



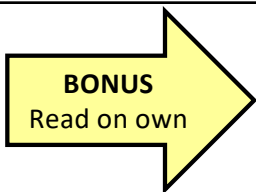
4. GLOVES

- Extend to cover wrist of isolation gown



DROPLET DONNING Sequence:

<http://www.cdc.gov/hai/pdfs/ppe/PPE-Sequence.pdf>



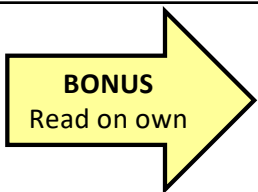
Precautions

Since COVID, consensus understanding that Aerosols/Droplets are on continuum rather than mutually exclusive categories. There is significant overlap and respiratory infections may have more or less of both categories

Precaution	What's involved
Airborne (A)	Negative pressure isolation with own washroom, mask (N95)
Droplet (D)	Mask, eye protection/face shield if respiratory illness
Contact (C)	Gown, gloves, dedicated equipment, single room if possible, own toilet if diarrhea

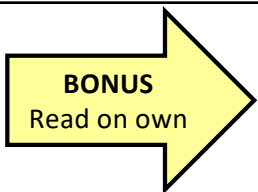
List of aerosol generating medical procedures (AGMP)

- Intubation/extubation, manual ventilation
- Open endotracheal/deep suction
- CPR (*not chest compressions, but airway manipulation)
- Bronchoscopy and bronchoalveolar lavage
- Tracheostomy/laryngoscopy
- Sputum induction
- Nebulized therapy (not MDI therapy)
- High flow nasal cannula (e.g. Venturi, Optiflow)
- Non-invasive positive-pressure ventilation (CPAP/BiPAP)
- Some dental procedures (drilling, ultrasonic scaling)



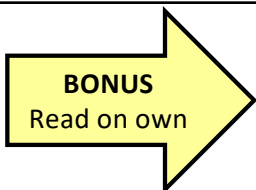
Precautions

Pathogen	Precautions	Duration of precautions
TB (pleural/laryngeal)	A	Until received 2 weeks of effective treatment and is clinically improving, and 3 consecutive negative sputums for AFB
Disseminated VZV	A + C	Until all lesions have crusted/dried
Primary or disseminated, extensive varicella	A + C	Until all lesions have crusted/dried
Measles	A	4d after start of rash
SARS-CoV 2/ COVID-19	D + C + respirator	10d following resolution of symptoms with minimum 24h symptom free if at home, 21d if immunocompromised/ICU admission/severe disease
Mumps	D	Until 5d after onset of parotitis
Meningococcus	D	Until 24h of effective rx has been received
VHF (including Ebola)	C + D	Until symptoms resolve



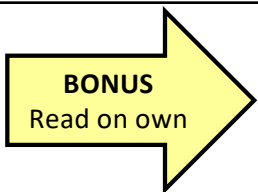
Precautions

Pathogen	Precautions	Duration of precautions
Invasive GAS (TSS/NF/pneumonia/meningitis)	D + C	Until 24h of appropriate abx received
C. difficile	C	Duration of symptoms
Disseminated/primary extensive HSV	C	Until all lesions dried/crusted
Diphtheria (pharyngeal)	D	Until 2 neg cultures from nose and throat
Seasonal influenza	D + C	Duration of symptoms
Norovirus	C	48h after resolution of illness
Scabies	C	Until 24h after initiation of appropriate therapy
Antibiotic resistant organisms (MRSA, VRE, ESBL, CRE, CPO)	C	As directed by ICP



Bites

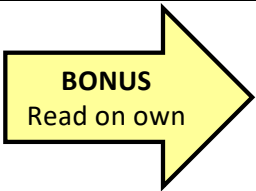
- Wound cleaning and irrigation
- Prophylactic or pre-emptive antibiotics
 - Amox/Clav for coverage of *Pasteurella multocida*, *Capnocytophaga canimorsus* in cat/dog bites, *Eikenella corrodens* in human bites
 - Consider in the following individuals:
 - Immunocompromised
 - Asplenic
 - Advanced liver disease
 - Pre-existing or resultant edema of affected area
 - Moderate to severe injury (esp. to hand or face)
 - Penetrating injuries to periosteum or joint capsule



Bites - Tetanus

- Tetanus prophylaxis in wound management

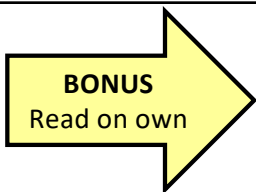
Hx of tetanus immunization	Clean, minor wounds		All other wounds	
	Toxoid-containing vaccine	Tlg	Toxoid-containing vaccine	Tlg
< 5 years since last booster and completed series	✗	✗	✗	✗
> 5 years but < 10 years since last booster and completed series	✗	✗	✓	✗
> 10 years since last booster and completed series	✓	✗	✓	✗
Unknown or incomplete series (< 3 doses) (e.g. immigrant populations)	✓	✗	✓	✓



Bites - Rabies

- Rabies post exposure prophylaxis
 - Immediate cleaning and flushing X 15min, avoid suturing
 - Call public health to assist in risk assessment
 - Ask about animal species, type of exposure, circumstances, behavior and vaccination status of animal, domestic vs stray, unprovoked vs provoked attack, location and severity of bite
 - If exposure to rabies is considered highly likely, administer post-exposure prophylaxis

	Rabies Immunoglobulin	Rabies Vaccine
Not vaccinated, immunocompetent	✓	✓ - 4 Doses
Not vaccinated, immunocompromised	✓	✓ - 5 Doses
Vaccinated	✗	✓ - 2 Doses

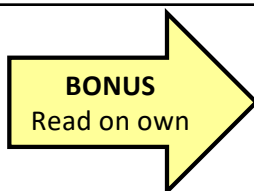


Bloodborne Pathogen Exposure Management

1. Source Patient
 - Status of HBV, HCV, and HIV.
 - If positive, treated or untreated? Controlled or not controlled?
 - If negative, risk factors (ie. MSM, multiple partners, PWID, etc.)
2. Exposure Factors
 - Type of fluid
 - High Risk: Blood, semen, vaginal secretions, visceral fluid (pleural, peritoneal, pericardial, amniotic)
 - Low Risk: Saliva, sputum, feces, urine, vomit, sweats, tears, nasal secretions
 - Type of injury
 - High Risk: Percutaneous (large hollow bore needles)
 - Moderate Risk: Splash to eyes, nose, mouth
 - Low Risk: Contact through non-intact skin
3. Host Patient
 - Immunization history

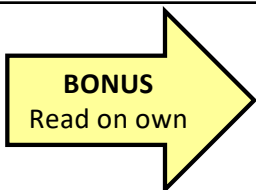
Source Patient	Exposed Patient
- p24 antigen-HIV antibody assay - Hepatitis B sAg - Hepatitis C Ab	- p24 antigen-HIV antibody assay - Hepatitis B sAg, sAb - Hepatitis C Ab - Pregnancy test - CBC, creatinine, ALT - STI testing - Repeat in 6 weeks and 3 months

	Time to positive serology	Time to positive NAAT
HIV	22 days	5-10 days
HCV	38-94 days	6-9 days
HBV	38-50 days	20-25 days



Bloodborne Pathogen Management

Infection	Risk of transmission if percutaneous exposure	Management/PEP
HIV	0.23% (up to 1.3%)	Truvada PLUS Dolutegravir (OR Raltegravir) x 28 days (within 72 hours of exposure)
Hepatitis B	6-30%	Immune (confirmed anti-HBs > 10mIU/mL): No further management Nonimmune (never vaccinated or anti-HBs < 10mIU/mL) or Unknown: Vaccine series +/- HBIG
Hepatitis C	3-10%	None. Repeat testing in 4-6 months.



Immunizations

References:

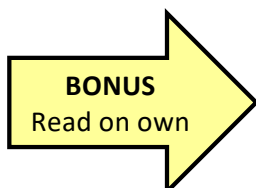
- Refer to National Advisory Committee on Immunization (NACI) Canadian Immunization Guide for further information:
 - <http://www.phac-aspc.gc.ca/naci-ccni/>
 - » Click on 'Canadian Immunization Guide' for key updates and information regarding specific vaccines

Great resource for patient education and up to date immunization schedule endorsed by PHAC, CMA, CFPC, AMMI Canada...:

[Immunize.ca](http://immunize.ca)

Vaccine Counselling

PHAC. Communicating effectively about immunization: Canadian Immunization Guide. 2016



- Ask patient if they are opening to discussing the vaccine, and what questions they may have
 - For vaccine hesitant patient, explore their reasons which may be multifactorial
 - Focus on what your patient's specific concerns are and not make assumptions
 - Respect differences in opinion, and adopt a patient-centred approach
- Discuss risks of vaccine preventable illness
- Discuss available vaccines, how they work, and their efficacy
- Be upfront about potential side effects

Principles of Effective Communication about Vaccines:

- Adopt a vaccine-recipient approach
- Respect differences of opinion
- Represent the risks and benefits of vaccines fairly and openly
- Clearly communicate current knowledge using an evidence-based approach

Benefits

- Effective at protecting against preventable diseases
- Safe
- Community/herd immunity indirectly protects those who cannot be vaccinated (newborns, elderly, immunocompromised)
- Cost savings to individual, healthcare system and society in prevention of illness

Risks

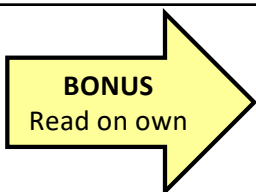
- Adverse reactions, generally minor and resolve quickly (fever, local injection reaction,
- Serious adverse reactions are rare (anaphylaxis, severe asthma exacerbation)
- Guillain-Barre syndrome within 6 weeks of immunization (influenza, tetanus toxoid, rare)

Types of Vaccines 101

BONUS
Read on own

Non-live (may also be called
inactivated): Cannot cause disease

- **Whole Vaccines:**
 - **Live Attenuated:** weakened live pathogen. Caution in immunocompromise (MMR, Varicella, Rotavirus, Immavune)
 - **Inactivated:** killed pathogen, cannot cause disease but often requires multiple doses. (Polio, Flu, Hep A)
- **Component Vaccines:** Uses part of pathogen. Often require repeat doses and adjuvants
 - **Subunit Vaccine:** Uses antigenic component of vaccine
 - **Protein:** Uses antigenic protein
 - **Polysaccharide:** Uses polysaccharide from bacteria. Less immune response (Pneumovax)
 - **Conjugate:** Polysaccharide attached to protein, to produce longer lasting immunity to polysaccharide (Prennar, Men ACYW)
 - **Toxoid:** Inactivated toxins produced by bacteria (Tetanus, Diphtheria)
 - **mRNA/DNA:** Viral RNA/DNA which is translated into protein in human cells (COVID Pfizer/Moderna)
 - **Recombinant Vaccines:** Insertion of viral gene into yeast/virus cell to produce pure antigen (Hep B, HPV)
- **Viral Vector (replicating or non-replicating):** Non-pathogenic viral vector carries gene for antigen, and produces protein inside body (COVID AstraZeneca)
- **Virus-like particles:** Molecules that closely resemble viruses (COVID Covifenz)



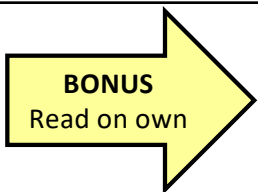
Adult Immunizations

Canadian Immunization Guide (PHAC, 2024)

Vaccine	Age				
	18-26 yrs	27-49 yrs	50-59 yrs	60 yrs	≥ 65 yrs
Td	✓ Booster q10 years				
PCV-20/PCV-21					✓
Shingrix			(✓)	✓	
Inf	✓ Annually				
HPV	✓	→			
Pertussis	1 dose as an adult and during each pregnancy				
COVID-19	Fall 2025: LP.8.1 mRNA if >6m from last vaccine or infection				
RSV	ABRYSVO (>60y or if pregnant 32-36 wks) or AREXVY (>60y) Note in Ontario publicly funded RSV vaccine for adults >60y and high risk (e.g. hematopoietic SCT or Solid Transplant or dialysis or homeless or First Nations or Long Term Care/Retirement living.				

Recommended immunizations, adults (18 years of age and older), previously immunized with primary series for diphtheria, tetanus, pertussis, polio, Hib, measles, mumps, rubella, meningococcal prior to age 24

- VZV – Shingrix (recombinant) for all over age 50 if no contraindication, consider Zostavax (live) if Shingrix contraindicated or unavailable
- HPV – Recommended for males and females ages 9 to 26 (and older)
- HepA and HepB – If at risk developing it



Adult Immunizations – Specific Populations

Vaccine	Examples of populations at risk	Schedule
Hib	Asplenia/hyposplenism, HIV, malignant hematologic disorders, PID, SOT, cochlear implant, HSCT	1 dose regardless of prior immunization hx
Meningo	Asplenia/hyposplenism, HIV, complement deficiency Travelers to meningitis belt, pilgrims to Hajj, military personnel, lab workers, close contacts of case	Men-C-ACYW X 2 doses, 8 wks apart; consider 4CMenB X 2 doses, at least 4 wks apart
Pneumo	Age >65, congenital deficiencies of the immune system, immunocompromising conditions or immunosuppressive therapy, HIV infection, HSCT recipients, active malignant neoplasms, recipients of solid organ transplants, chronic kidney disease, chronic liver disease, functional asplenia, CSF leak, chronic heart disease, cochlear implants, diabetes mellitus, chronic lung disease.	1 dose Pneu-C-20 or Pneu-C-21

BONUS
Read on own



Choosing Wisely and ID

- Don't routinely prescribe IV forms of highly bioavailable antimicrobial agents for patients who can reliably take and absorb oral meds (ciprofloxacin, metronidazole, fluconazole)
- Don't prescribe alternate second-line antimicrobials to patients reporting non-severe reactions to PCN when beta-lactams are recommended first-line therapy
- Don't routinely repeat CD4 counts in patients with HIV and undetectable VL > 2 years and CD4 > 500 unless virologic failure or intermittent OI
- Don't routinely repeat imaging in patients with OM demonstrating clinical improvement following adequate antimicrobial Rx
- Don't prescribe aminoglycosides for synergy to patients with bacteremia or NV IE caused by *S. aureus*

Bonus MCQ 1 + 2

Bonus 1 Endocarditis

40 year old male with *Strep. mitis* endocarditis on a bicuspid aortic valve. Echo shows mild AI and small vegetations on both cusps. Treated with ceftriaxone. 10 days after initiating therapy he develops a cold, ischaemic pale leg. Symptoms last 6 hours then resolve. Next step:

- A. Change antibiotics to vancomycin
- B. CT head
- C. Consult CV surgery for surgical assessment
- D. IV heparin followed by warfarin
- E. Continue present management

Bonus 2 Diarrhea

A 30 year old female presents with diarrhea and abdominal pain. She has had 5 days of fever. She returned from Cambodia 9 days ago. On examination her temperature is 39.6 and she has a palpable spleen tip. She also has a rash consisting of 1-4mm erythematous macules on her trunk. Which antibiotics would you give?

- A. Ampicillin
- B. No Antibiotics
- C. Metronidazole
- D. Ceftriaxone

MCQ Bonus 3 + 4

BONUS 3 - HIV

68 y M from Saudi Arabia living with HIV admitted with PJP pneumonia confirmed by bronchoscopy. ABG shows PaO₂ 62. He has a severe allergy to sulfa reported. In addition to prednisone, management should include:

- a. TMP-SMX 15mg/kg IV
- b. IV foscarnet
- c. IV pentamidine
- d. PO dapsone
- e. Clindamycin and Primaquine

BONUS 4 – STI

A 28M develops 2 weeks of fever, bloody mucoid diarrhea, and tenesmus. A colonoscopy demonstrates inflamed and friable tissue at the level of the rectum. Tissue biopsies show epithelioid granulomata showing severe central necrosis and a urine NAAT is negative for Chlamydia. Which medication do you prescribe?

- a. Metronidazole
- b. Doxycycline
- c. Acyclovir
- d. Prednisone

MCQ Bonus 5 + 6

BONUS 5 – STI

You are asked to see a 22-year-old female college student with diffuse pustular skin lesions, and migratory arthralgias. Last week, she was seen in the ED for a swollen knee which has now resolved. What is the best way to make the diagnosis?

- A. Wound swab or lesions
- B. Blood cultures x 2
- C. HIV serology
- D. Urine, cervical, rectal and pharynx testing for C+G

BONUS 6 – STI

You have been referred a 78M patient for positive syphilis serology (RPR 1:32), ordered in the context of work-up for dementia. They do not have a previous history of syphilis have never been previously tested, and are asymptomatic aside from some intermittent ringing in their ears. They have already been given one dose of benzathine pen G by their family physician last week. The remainder of screening of STIs is negative. What further work-up do you recommend?

- A. No further work-up or treatment.
- B. Continue benzathine pen G 2.4 million units IM x 2 more doses 1 week apart, for total 3 doses
- C. Repeat syphilis serology now to assess RPR titre for decrease to further guide treatment
- D. Follow-up in 3 months with repeat serology at that point. Repeat penicillin dose then if titre has not decreased 3-fold
- E. Lumbar puncture

BONUS MCQ 7+ 8

BONUS 7 - Malaria

A young man presents to the ED with fevers and confusion, after returning from a trip to visit his relatives in Nigeria for summer break. He is icteric, and bloodwork reveals a Hgb 52, Plat 44, Creat 150, lactate 4.5, INR 3.3. He has a blood smear that confirms *P. falciparum* with 8% parasitemia. Which of the following is part of your management.

- A. Admit to hospital ward
- B. Start double-dose atovaquone-proguanil
- C. Lumbar puncture
- D. IV artesunate

BONUS 8 - Lyme

You have seen a fisherman in southern Nova Scotia who presents with a large rash on their leg with central clearing. They do not feel unwell, but do not know how long it has been there. There are no other rashes, and you do not see a tick anywhere. They have no joint pain, or neurologic findings. What is your treatment approach?

- A. Lyme serology, and start treatment if positive
- B. Doxycycline 200mg PO x 1 now, Lyme serology and reassess in 1 week.
- C. Lyme serology, doxycycline 100mg PO BID x 10d
- D. Doxycycline 100mg PO BID x 10d
- E. Lyme serology, Cephalexin 500mg PO BID x 7 d for potential cellulitis

Bonus 9 – COVID-19 2024

A 87 year old woman with 4 previous mRNA vaccine doses, but no booster in the past 12 months, presents with cough and fever. She is found to be COVID positive. Her oxygen saturation is 97% on room air. Her medical history includes T2DM, HTN and chronic kidney disease (eGFR 50), and her current medications are metformin and ramipril. She is fatigued, but comfortable going home. What do you recommend?

- A. Amoxicillin-Clavulanate for 5 days
- B. Remdesivir for 3 days
- C. Nirmatrelvir/Ritonavir x 5 days
- D. No treatment

BONUS MCQ 10 + 11

BONUS MCQ 10 – CNS Infection

You have just confirmed a case of N. meningitidis meningitis in a 21 year old university student. What should you NOT do?

- A. Call public health at 3AM to alert them.
- B. Give prophylaxis to all ICU staff caring for this patient as he was not on isolation initially.
- C. Place patient on droplet precautions as still febrile.
- D. Give vaccine and prophylaxis for his university roommate.

BONUS MCQ 11 – Resistant Organisms

A 67 year old man is intubated in the ICU after a traumatic brain injury 2 weeks ago. Over the past 24 hours he has required increasing FiO₂ and pressure support. A sputum sample is taken and grows *Enterobacter cloacae*. The report indicates it is susceptible to all beta-lactams. What antibiotic do you treat with?

- A. Vancomycin
- B. Ertapenem
- C. Piperacillin-Tazobactam
- D. Ceftriaxone

BONUS MCQ 12 + 13

BONUS MCQ 12 – Travel

You are seeing a 27 year old man on the ward with pneumonia. He returned from Thailand 8 days ago, where he was staying at a hostel. He has a left lobe infiltrate on CXR. He has been on Ceftriaxone for 4 days without improvement. What pathogen do you suspect?

- A. Legionella
- B. S. pneumoniae
- C. Mycoplasma
- D. Tuberculosis

BONUS MCQ 13 – Immunocompromise

A 60 year old woman with AML was admitted with febrile neutropenia and treated with Meropenem and vancomycin. She is also on Ganciclovir due to a positive blood CMV PCR. 1 week into therapy she has a repeat fever and is found to have candidemia and is started on fluconazole for 2 weeks. She initially improves. She has been neutropenic for the past month, but her neutrophils are now beginning to improve. 1 week later, she becomes febrile again. Bloodwork shows Neutrophils 1.5, ALT 100, ALP 400, Bilirubin 20. What is the cause of her fever?

- A. CMV viremia
- B. Hepatic Abscess
- C. Drug Fever
- D. Hepatosplenic Candidiasis

Bonus MCQ 14+15

BONUS MCQ 14: Tb + HIV

You have just diagnosed active pulmonary TB. In which patient would the likelihood of MDR-TB be lowest?

- A. Patient with thoracotomy 20 years ago for TB
- B. Patient with well controlled HIV
- C. Patient from a northern First Nations community with recent contact with MDR TB patient
- D. Patient from South Africa

MCQ 15: GU Infections

You are seeing a post-menopausal female patient in the ED. She has fever, flank pain and nausea. You suspect pyelonephritis. She had an episode of cystitis one month ago secondary to ESBL E coli (S FQ, TMP-SMX, nitrofurantoin, fosfomycin). It was treated with nitrofurantoin. After sending off supportive testing, how do you initially treat?

- A. Fosfomycin PO
- B. TMP-SMX PO
- C. Moxifloxacin IV
- D. Ertapenem IV

Bonus 16- IPAC - 2024

You are seeing a 25 year old man in the ED with new fever, cough and hypoxia without any abnormalities on CT chest; it is not influenza season. He was placed on droplet/contact precautions by the ED for suspected COVID and an NP swab was sent. During your interview, he mentions that he works at a poultry farm. What should you do next?

- A. Contact IPAC and Public Health
- B. Start Ceftriaxone
- C. Contact Respiriology for a BAL
- D. Prophylaxis for all household contacts

MCQ Bonus 17 + 18

BONUS 17 – HIV

A young man is seen in clinic for a new diagnosis of HIV. On physical examination, you do not note any abnormalities, but he points out some “bruises” on his arms that appeared months ago and have not healed. There are three, all less than 1cm in diameter. On bloodwork, his CD4 count is 47, viral load is pending. Which treatment will you arrange for these lesions?

- A. Intralesional chemotherapy
- B. Surgical excision
- C. Start atovaquone
- D. Start BIC/FTC/TAF

BONUS 18 – Intraabdominal Infections

An 67 year old man undergoes a colectomy for perforation secondary to colon cancer. He is placed on ceftriaxone and metronidazole. He develops a fever on post op day 2, but is otherwise stable. Abdominal ultrasound reveals a 5 cm abdominal abscess. What is the next step in management?

- A. Laparotomy
- B. Percutaneous Drainage
- C. Change antibiotics to Piperacillin-Tazobactam
- D. Add Fluconazole

BONUS MCQ

BONUS MCQ 19 - endocarditis

You see a 72yF with bioprosthetic aortic valve for shortness of breath on exertion. She reports some night sweats for past few weeks and recalls it started after her tooth became infected a month ago, and she had a delay until could be seen by dentist. On your physical examination the patient is stable, 37.8C, BP 110/60, HR 80. She has splinter hemorrhages and diastolic blowing murmur in LLSB.

Which of the following is not appropriate initial management?

- a) Blood cultures
- b) Transthoracic echocardiogram
- c) No empiric antibiotics
- d) IV Ceftriaxone and Ampicillin

BONUS MCQ 20 – Bone/joint

You are seeing a patient in follow-up for congestive heart failure and hyperglycemia after a recent hospital admission. They ask you to assess an ulcer on their foot, as they are worried that it might have become infected since their last admission to hospital. Which of the following is most indicative of infection?

- A. Bad odour
- B. Increased redness
- C. Purulence at the site
- D. Increased pain

MCQ Bonus 21 + 22

BONUS MCQ 21 - Travel

You are asked to see a 22 year old woman in emerge who had returned from Vietnam 22 days ago. They present with 3 days of fever, headache, maculopapular rash and thrombocytopenia. Which of the following is LEAST likely?

- A. Dengue
- B. Malaria
- C. Typhoid
- D. HIV
- E. COVID-19

BONUS MCQ 22 - Parasites

You are seeing a 53 year old man with who returned 4 weeks ago from visiting relatives in India for 6 months. For the last two weeks he has been experiencing fevers and night sweats and has a dull pain in his right upper quadrant. You notice some RUQ tenderness on exam and order an ultrasound which shows well-defined hypoechoic abscess in the posterior right lobe. Which amoeba causes amebic liver abscesses?

- A. Naegleria fowleri
- B. Acanthamoeba
- C. Entamoeba histolytica
- D. Entamoeba dispar

MCQ Bonus 23 + 24

BONUS MCQ 23 – Fungal Infections

A 75F known for a past medical history of T2DM was recently treated with a course of dexamethasone during an inpatient hospitalization for COVID pneumonia. Patient presents two weeks post-discharge with unilateral facial pain. You examine her nasal septum and there is a black necrotic eschar. What antimicrobial therapy do you initiate?

- a. Flucytosine
- b. Amphotericin B
- c. Voriconazole
- d. Caspofungin

BONUS MCQ 24 – Infectious Diarrhea

A 35M presents to the ED with a 5 day history of bloody diarrhea after returning from India in recent days. Symptoms started during the flight home and have progressed. He is hypotensive with a fever. In addition to IV fluid resuscitation and supportive management which medication will you prescribe?

- a. Azithromycin
- b. Ciprofloxacin
- c. Septra
- d. Loperamide

MCQ Bonus 25 + 26

BONUS MCQ 25 – STI

A 28M develops 2 weeks of fever, bloody mucoid diarrhea, and tenesmus. A colonoscopy demonstrates inflamed and friable tissue at the level of the rectum. Tissue biopsies show epithelioid granulomata showing severe central necrosis and a urine NAAT is negative for Chlamydia.

Which medication do you prescribe?

- a. Metronidazole
- b. Doxycycline
- c. Acyclovir
- d. Prednisone

BONUS MCQ 26 - TB

38 year old ER nurse now finds that her annual repeat screening TST is newly positive. She does not have any recent history of fever, night sweats, weight loss, or hemoptysis. A chest X-ray is normal and not concerning for TB disease (active TB). An induced sputum does not demonstrate any AFB and mycobacterial culture is later finalized as negative. According to Canadian guidelines, which of the following regimens is **NOT** first-line for treatment of this patient?

- a. Rifapentine + Isoniazid for three months
- b. Isoniazid for 9 months
- c. Rifampin for 4 months
- d. All three are appropriate first-line options.

MCQ Bonus 27 + 28 2026

BONUS MCQ 27 – UTI/Microbiology

Patient presents with ED with complicated UTI, blood and urine cultures are positive for *Citrobacter koseri*, which antibiotic is appropriate to start empirically?

- a) Ceftriaxone
- b) Ertapenem
- c) Meropenem
- d) Nitrofurantoin

BONUS MCQ 28 – Toxic Shock

Patient presents with group A streptococcal toxic shock syndrome. In addition to hypotension and a positive culture for group A streptococcus from a sterile site, which of the following is **NOT** a diagnostic criteria for this condition:

- a. Creatinine over 177 $\mu\text{mol/L}$
- b. ARDS
- c. Platelets under 100
- d. Hemoglobin under 70

MCQ Bonus 29 + 30

BONUS MCQ 29 – IPAC

Which of the following infections would **NOT** require the use of droplet precautions in an inpatient setting:

- a) Meningitis with CSF positive for *Neisseria meningitidis*
- b) Group A streptococcus pneumonia with bacteremia
- c) *Legionella* pneumonia
- d) Pharyngeal infection with *Corynebacterium diphtheriae*

BONUS MCQ 30 – Infective Endocarditis

54F is admitted for infective endocarditis with blood cultures positive for *E. faecalis*. Blood cultures cleared after 2 days once appropriate antibiotics are started. A TEE demonstrates a 7-millimetre mobile vegetation on the mitral valve. Two weeks later she remains in hospital. Of the listed complications below, which is **NOT** an indication to reassess the need for early surgery according to ACC/AHA guidelines?

- a) Repeat echo shows new MV annular abscess
- b) New episode of fever, and blood cultures are again positive for *E. faecalis*
- c) She develops complete heart block identified on ECG
- d) There is a new murmur, and echo shows new asymptomatic valvular dysfunction
- e) New evidence of a renal infarct secondary to septic embolus

BONUS MCQ Answers

- 1) C** - This is one of the indications for consideration of surgical approach (**emboli despite antibiotic therapy**). Know these for your exam. No indication to change antibiotics (strep universally susceptible).
- 2) D** - Typhoid is common in southeast Asia. Presentation includes fever, "rose spots", abdominal pain and can develop hepatosplenomegaly. Constipation is most common presentation but can present with diarrhea. Preferred treatment, particularly in inpatients, is ceftriaxone.
- 3) C** - He is Sulfa Allergic so TMP-SMX is off the table. For second line – clinda/primaquine usually preferred but being from Saudi Arabia higher chance this man has G6PD deficiency so would recommend PENTAMIDINE instead.
- 4) B** – Patient is presenting with LGV-infection with rectal involvement. A negative urine NAAT does not mean that there is not Chlamydia elsewhere, and when patients are being tested they should have oral and rectal swabs together with urine testing
- 5) D** - Clinically this is suspicious for disseminated gonococcal infection. Can try to swab the skin lesions, but this may be lower yield than performing NAAT PCR at all sites of sexual contact (cervical, urine, +/- rectum/pharynx). HIV serology would also be indicated for risk factor screening. Yield of blood cultures is low, despite likely hematogenous spreading of bacteria.
- 6) E** - At minimum this is late latent syphilis of unknown origin, but sx of dementia, + ringing in ears could be findings of neurosyphilis/otic syphilis. This requires LP, and treatment with IV penicillin x 14d. If this was negative, then continue treatment for late latent as in option B but need to rule out neuro first. Repeating serology now is too early, but goal is 4-fold drop within 6 months (if latent) or 1 year (if neuro).
- 7) D** - This patient has severe malaria (anemia, potentially DIC, high parasite burden assuming he is non-immune). He requires admission to ICU for IV artesunate. His current coagulopathy precludes safe LP.
- 8) D** - Nova Scotia has +++ Lyme disease. This sounds like uncomplicated Lyme, or erythema migrans. Serology is NOT needed to diagnose, and should not delay treatment. Should be treated up front with doxycycline x 10 days as long as no evidence of disseminated disease
- 9) C** - In a patient over 70 with >3 high risk comorbidities presenting with mild COVID, treatment can be offered regardless of vaccine status (although benefit is not totally clear). Due to her CKD she will need a dose reduction for Nirmatrelvir/Ritonavir; however she has no medication interactions. Remdesivir is IV and not convenient for outpatients.



BONUS MCQ Answers

- 10) B-** In this case B is the best answer because prophylaxis is not warranted for all ICU staff, only those who have had intensive unprotected contact (eg. intubating, resuscitating, closely examining oropharynx without a mask). The others you should do.
- 11) B** - Enterobacter is a high risk ECK/AmpC organism that may induce resistance to penicillin and cephalosporins that isn't present at the time of initial culture results. Preferred treatment is Carbapenem or non-beta lactam (however Vancomycin does not cover gram negative organisms)
- 12) A** - Legionella outbreaks occasionally occur in hotels/hostels in water supply. There have previously been outbreaks in travelers to Thailand. It would not respond to Ceftriaxone. S. pneumoniae expect response to Ceftriaxone. Mycoplasma rarely can cause mild pneumonia with bilateral infiltrates, not likely to be severe. TB uncommon in travelers, uncommon to present with bilateral infiltrates.
- 13) D** - Hepatosplenic Candidiasis typically occurs in patients with heme malignancy and prolonged neutropenia. Candida gets into the bloodstream and seeds the liver/spleen. Lesions are often not visible while neutropenic, but emerge once the neutrophil count recovers. May be accompanied by an elevated ALP. Suspect in someone who either is not improving on antifungals or improves then worsens again especially after neutrophil recovery. Drug fever could also cause fever and elevated ALP, but overall pattern consistent with hepatosplenic candidiasis
- 14) B** is the best answer based on the guidelines as known risk factors for MDR include previously treated TB (anti-TB medications would have been used 20 years ago), known contact with an MDR case and being born in an endemic country .
- 15) D-** Ertapenem. Moxifloxacin has a lower concentration in the urine so not approved for this indication (Pick levofloxacin or ciprofloxacin for urinary infections instead).
- 16) A** - This exposure is concerning for a possible H5N1. Contact IPAC and public health for further guidance. The patient is already on contact/droplet precautions however some hospitals will request airborne precautions. While you could consider empiric treatment, prophylaxis for contacts would not be done unless the diagnosis confirmed
- 17) D - D-** Sounds like Kaposi's sarcoma. Usual treatment is initiation of ART, and follow up as this generally regresses with treatment. Intralesional chemotherapy can be considered for large lesions, especially if not improved on ART.
- 18) B** - Percutaneous drainage is mainstay of treatment for abdominal abscess. For community acquired abdominal infections, Ceftriaxone and Metronidazole would be reasonable coverage (no indication he is on chemo in question), so no urgency to switch. Empiric candida coverage not required after only 2 days.

BONUS MCQ Answers

19) D - This patient does not yet have a diagnosis of endocarditis (does not meet Duke Criteria based on provided info alone – 3 minor criteria). If a patient is stable, empiric antibiotic should be held until A) diagnosis is made based on Duke criteria and B) at least 3 sets of blood cultures have been obtained. Generally continue to hold antibiotics until blood cultures are resulted and either organism is obtained or culture negative endocarditis is diagnosed. For patient with suspected endocarditis who are a) acutely ill (e.g. septic) or b) show high risk signs (e.g. concern for embolization) and no organism yet obtained, empiric therapy should be started after initial blood cultures taken, following the guidelines for culture negative endocarditis.

Based on IDSA: 1. Acute native valve endocarditis: *S. aureus*, B-hemolytic streptococci, and aerobic GNB

2. Subacute native valve endocarditis: *S. aureus*, VGS, HACEK and enterococci

3. Prosthetic Valve Endocarditis <1 year from placement: staphylococci, enterococci and aerobic GNB

4. Prosthetic Valve Endocarditis >1 from placement: staphylococci, VGS, enterococci

*Vancomycin + Ceftriaxone is reasonable coverage for all of these

20) D - JAMA RCE: Pain in chronic wound (LR 11-20), Foul odor (LR 1-3), Purulence, exudate, erythema, warmth, and edema (LR < 1.0)

21) A. DENGUE - Incubation periods critical. Symptom onset > 14 days after travel essentially rules out dengue

22) C - The travel history and appearance is suggestive of an amebic liver abscess. Amebic liver abscesses are caused by *Entamoeba histolytica*. *Entamoeba dispar* is a non-pathogenic species that resembles *E. histolytica* on microscopy. *Naegleria* and *Acanthamoeba* primarily cause CNS related disease. *Naegleria* is also not found in Canada. The incubation period is 5 days and death usually follows 5 days after symptom onset, so it is unlikely that a case even in a traveller would occur be seen in Canada.

23) B – This is a typical presentation for Mucormycosis in a patient that has multiple risk factors for infection with T2DM and recent steroid use.

Surgical debridement is urgent, and amphotericin is the appropriate first-line agent for initial therapy, which can later be stepped down to Posaconazole or Isavuconazole. Flucytosine, voriconazole, and capsosungin do not have activity against Mucor.

BONUS MCQ Answers

- 24) A** - For real-world practice, dividing the globe between susceptible and resistant to fluoroquinolones as included in the section on infectious diarrhea is inaccurate as there are countries on the left side that have high rates of resistance and there are countries on the right side that have very low rates due to a history of antibiotic stewardship. For the exam, memorizing rates of resistance by country is not reasonable, so if asked about resistance this over-simplification can be used.
- 25) B** - Patient is presenting with LGV-infection with rectal involvement. A negative urine NAAT does not mean that there is not Chlamydia elsewhere, and when patients are being tested they should have oral and rectal swabs together with urine testing. Treatment is doxycycline for 21 days.
- 26) B** - Although INH for 9 months was previously first line and remains first line in other national and international guidelines, the 2022 Canadian 8th TB standard relegated INH9 to second line because of its duration, not because of inferior efficacy.
- 27) A** – Remembering the new ECK acronym for high-risk AmpC organisms (which would typically require a carbapenem empirically, the C stands for *Citrobacter freundii* only. *Citrobacter koseri* lacks the AmpC gene completely, making ceftriaxone an acceptable empiric choice for a complicated UTI
- 28) D** – Remember the CLARK mnemonic (Coagulopathy, Liver enzymes, ARDS, Rash, Kidneys) to find that hemoglobin is not a criteria
- 29) C** – *Legionella* is not spread person-to-person, so routine precautions are all that is necessary
- 30) D** - Strictly speaking the correct answer is (d), as the guidelines specify that progressive valvular dysfunction WITH heart failure that is refractory to medical management is a Class I indication for early surgery. In practice however, even asymptomatic progressive valvular dysfunction should still be discussed by the multi-specialty heart valve team.